CHAPTER 09

Carboxylic Acid Derivatives

Content and rendering review packet

Original app content and actual Chromium screenshots. No textbook passages or textbook figures are embedded. Source labels and curriculum headings are not evidence of factual correctness.

Source snapshot SHA-256: e0ef507421b006b2ba207e7698d447b6380e1800534cada659708d80f8aeeb33

Review all claims, molecule graphs, lessons, exercises, hints and displayed states. Keys are author claims, not ground truth. Screenshots show static states with reduced motion: they do not prove animation timing or persistence behavior.

For every finding: item ID/page; severity (critical/major/minor); exact claim; derivation or reference evidence; proposed correction; confidence. List unreviewed/uncertain items. Model agreement is not scientific verification.

Coverage checklist

ch09:exercise:cyclic-derivative | cyclic-derivative:stage-1, cyclic-derivative:stage-2, cyclic-derivative:stage-3

ch09:exercise:acyl-order | acyl-order:stage-1, acyl-order:stage-2, acyl-order:stage-3

ch09:exercise:hydrolysis-products | hydrolysis-products:stage-1, hydrolysis-products:stage-2, hydrolysis-products:stage-3

ch09:lesson:families | families:rule, families:cue, families:trap, families:app, families:step-1, families:step-2, families:step-3

ch09:lesson:acyl-mechanism | acyl-mechanism:rule, acyl-mechanism:cue, acyl-mechanism:trap, acyl-mechanism:app, acyl-mechanism:step-1, acyl-mechanism:step-2, acyl-mechanism:step-3

ch09:lesson:derivative-reactions | derivative-reactions:rule, derivative-reactions:cue, derivative-reactions:trap, derivative-reactions:app, derivative-reactions:step-1, derivative-reactions:step-2, derivative-reactions:step-3

ch09:molecule:ester | ester

ch09:molecule:betalactam | betalactam

ch09:molecule:lactone | lactone

ch09:molecule:methylacetate | methylacetate

ch09:molecule:ester-intermediate | ester-intermediate

ch09:molecule:anhyd | anhyd

ch09:molecule:acetamide | acetamide

ch09:curriculum | scope-1, scope-2, scope-3

Curriculum scope

scope-1: 9.1 Amides, Esters, and Anhydrides

scope-2: 9.2 Reactivity Principles

scope-3: 9.3 Nucleophilic Acyl Substitution Reactions

Objective: Identify esters, amides, anhydrides, lactones, and lactams

Objective: Compare acyl substitution reactivity

Objective: Predict hydrolysis and transesterification outcomes

Supplied Kaplan curriculum: Chapter 9; section scope checked against local body and summary | organic_chemistry_kaplan/OEBPS/chapter09b.xhtml

Nucleophilic acyl substitution: OpenStax 21.2, inspected | https://openstax.org/books/organic-chemistry/pages/21-2-nucleophilic-acyl-substitution-reactions

Carboxylic acid reactions and esterification: OpenStax 21.3, inspected | https://openstax.org/books/organic-chemistry/pages/21-3-reactions-of-carboxylic-acids

Acid derivative naming: OpenStax 21.1, inspected | https://openstax.org/books/organic-chemistry/pages/21-1-naming-carboxylic-acid-derivatives

No source passage is included. Request relevant authoritative excerpts when a claim cannot be checked by derivation.

families | Recognize acid derivatives

Read the atom directly attached to acyl carbon

families:rule | Core Rules

Esters have C(=O)–OR; amides have C(=O)–NR2; anhydrides have C(=O)–O–C(=O). Lactones are cyclic esters and lactams are cyclic amides. A beta-lactam contains a four-membered ring.

families:cue | Memory Cues

Acyl C → O–C: ester. Acyl C → N: amide. Acyl C → O → acyl C: anhydride. For esters, name the O-alkyl group then the acid-derived -oate. Amides use -amide, with N- prefixes for N substituents. Anhydrides use the acid name(s) plus anhydride; list both acids for a mixed anhydride. In lactam/lactone shorthand, beta = four ring atoms, gamma = five, delta = six; count the ring heteroatom and carbonyl carbon, excluding the exocyclic oxygen.

families:trap | Common Mistakes / MCAT Traps

An ether has no adjacent carbonyl. An N-substituted amide is still an amide even if it has no N–H bond. Ring size names count heteroatoms as ring members. Ammonia, primary and secondary amines can give ordinary neutral amides by acylation and N–H loss. Tertiary amines lack that N–H: they can act as bases or form acylammonium intermediates, rather than the analogous neutral amide by simple acylation.

families:app | MCAT Applications

Triacylglycerols contain ester links; proteins contain amide links. Beta-lactam antibiotics exploit a strained amide structure.

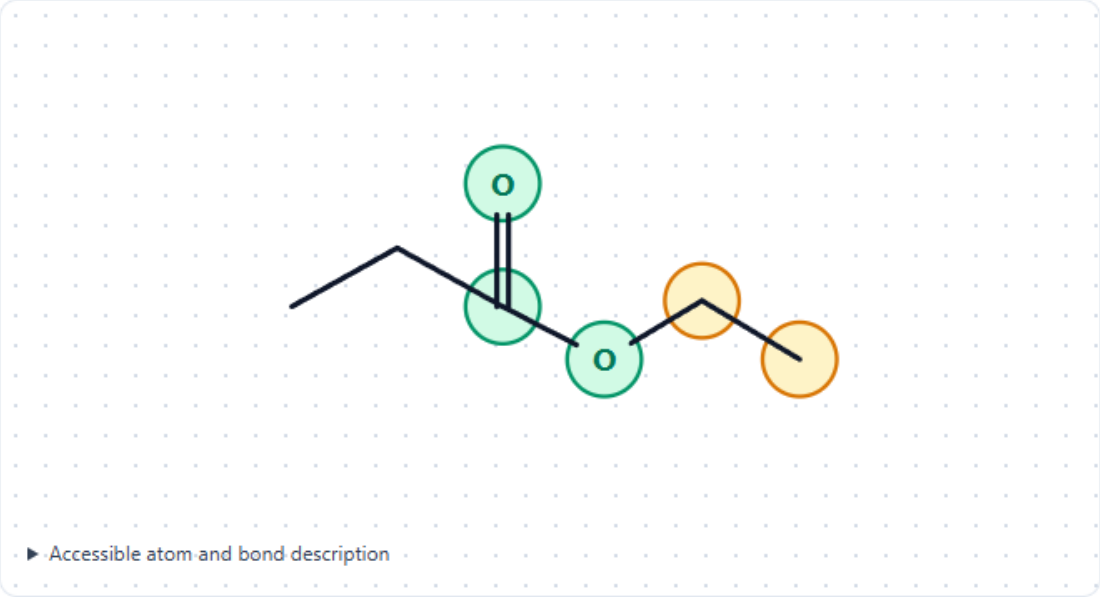
families:step-1 | 1. Acyclic ester

Ethyl propanoate connects the acyl carbon to oxygen. The alkoxy carbon chain lies beyond that oxygen. An amide instead bonds the acyl carbon directly to nitrogen.

Displayed molecule ID: ester

{"highlights": {"prin": [2, 3, 4], "sub": [5, 6]}, "revealName": true}

Screenshot: families-step-1



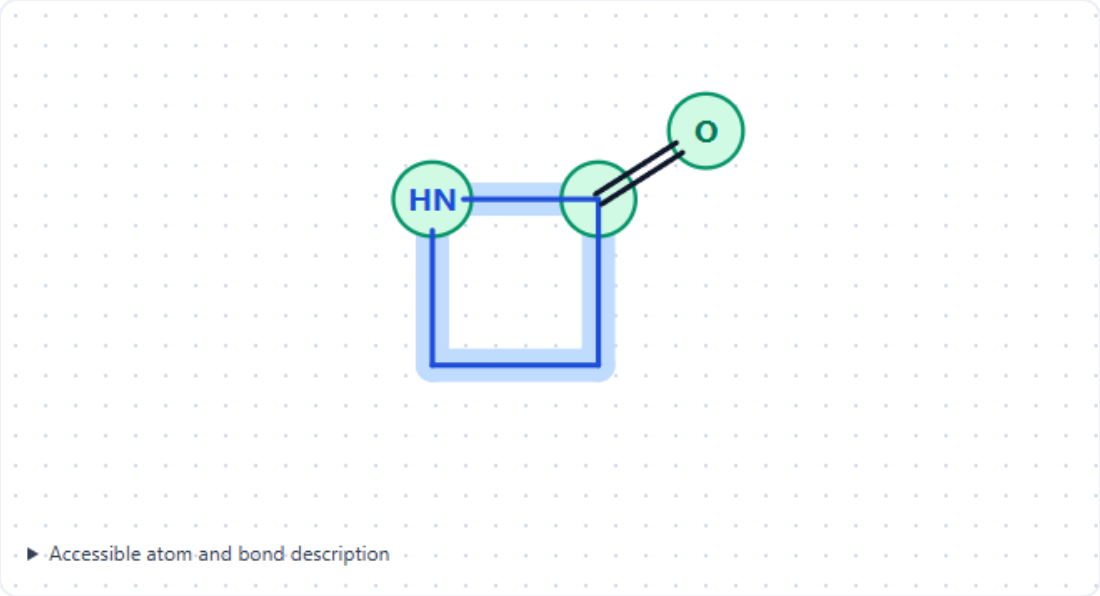
families:step-2 | 2. Cyclic amide

The beta-lactam has four ring atoms, one of them N. The carbonyl carbon is part of the ring and the oxygen is outside it.

Displayed molecule ID: betalactam

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Screenshot: families-step-2



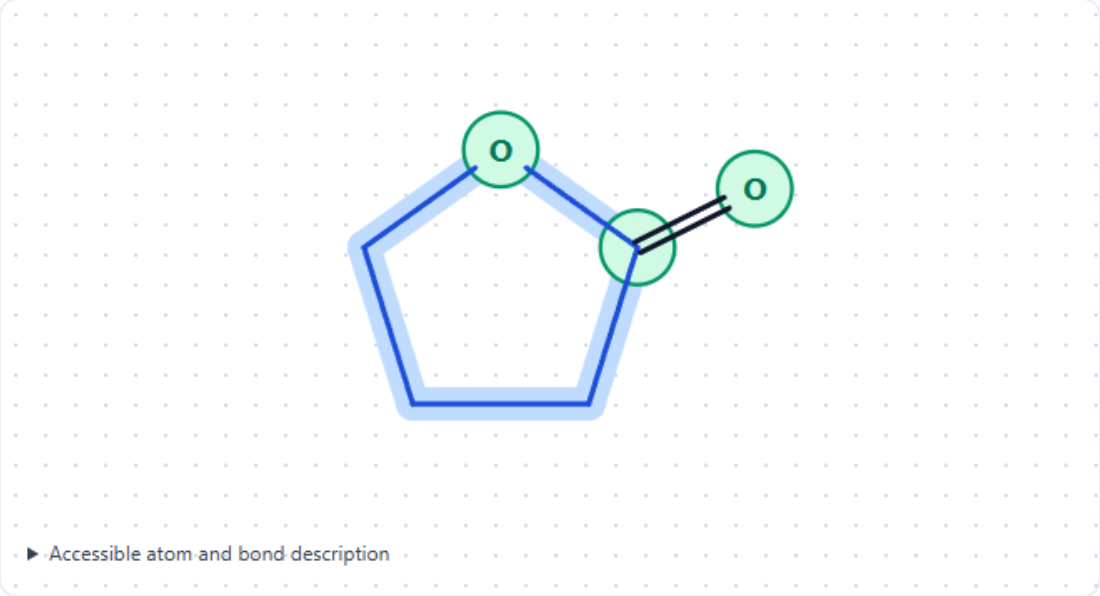
families:step-3 | 3. Cyclic ester

A lactone contains an ester linkage within a ring. This five-membered lactone has one ring oxygen and four ring carbons.

Displayed molecule ID: lactone

{"highlights": {"path": [0, 1, 2, 3, 4], "prin": [0, 1, 5]}, "revealName": true}

Screenshot: families-step-3



acyl-mechanism | Acyl substitution and relative reactivity

Addition → tetrahedral intermediate → elimination

acyl-mechanism:rule | Core Rules

A nucleophile adds to an acyl carbon and shifts the pi pair to O. Collapse of the tetrahedral intermediate reforms C=O and expels a leaving group. For comparable unactivated derivatives, a useful order is acid chloride > anhydride > ester > amide. Leaving-group basicity and resonance donation both matter. Inductive withdrawal transmits through sigma bonds and can increase acyl-carbon electrophilicity. Steric crowding can hinder nucleophile approach. These effects compete with leaving-group ability, resonance and catalysis; a single naming-priority or conjugation shortcut cannot rank every reaction.

acyl-mechanism:cue | Memory Cues

Addition–elimination replaces the acyl substituent while retaining a carbonyl.

acyl-mechanism:trap | Common Mistakes / MCAT Traps

Aldehydes and ketones normally add rather than substitute because hydride or an ordinary carbanion would be a poor leaving group. Conjugation means adjacent aligned p orbitals delocalize electrons; it can stabilize a reactant, intermediate or transition state. A rate depends on the activation free-energy difference, so stabilization alone does not prove a faster reaction. Enones also offer beta attack (1,4-addition), which differs from direct carbonyl attack (1,2-addition). Heteroatom lone-pair donation into C=O, especially from amide N, reduces carbonyl electrophilicity. The effect of conjugation requires the actual substrate, nucleophile, site and conditions.

acyl-mechanism:app | MCAT Applications

Amide resonance makes peptide bonds relatively resistant to uncatalyzed hydrolysis. Strained beta-lactams are more reactive than ordinary unstrained amides. Small rings can have angle strain and torsional strain. Beta-lactam geometry can reduce optimal amide orbital overlap and hydrolysis can relieve strain; the extent of nitrogen pyramidalization and resonance loss depends on ring fusion/substitution. The displayed unfused four-membered connectivity diagram is not a 3D bond-angle model.

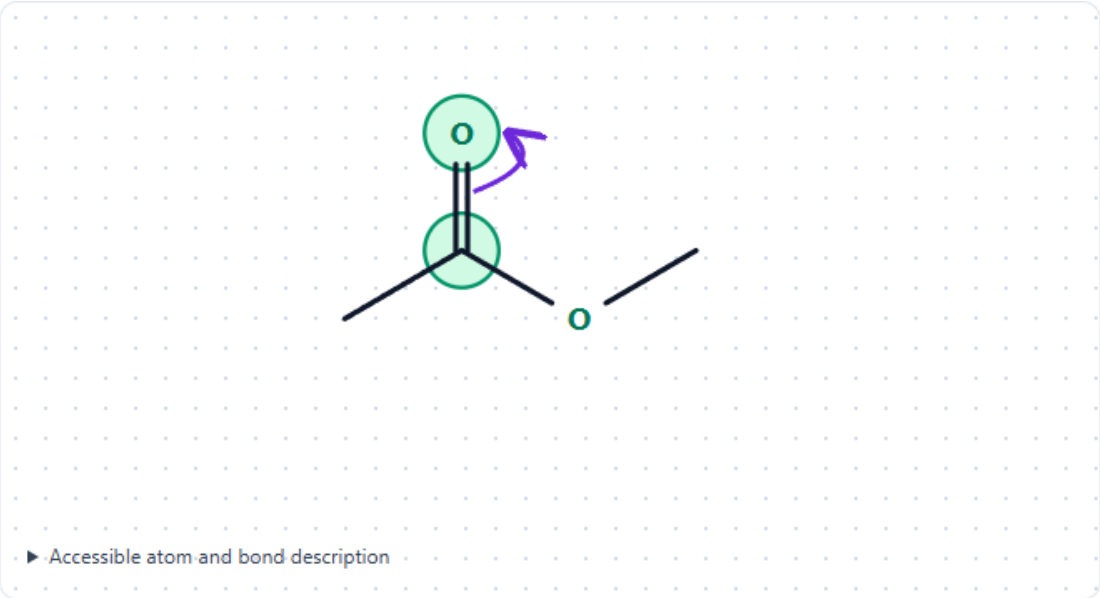
acyl-mechanism:step-1 | 1. Add to the carbonyl

Hydroxide attacks the ester carbonyl carbon while the pi pair moves to oxygen. The arrow shows that pi-pair movement, not the entire mechanism.

Displayed molecule ID: methylacetate

{"highlights": {"prin": [1, 2], "flow": [{"fromBond": 1, "toAtom": 2}]}, "revealName": false}

Screenshot: acyl-mechanism-step-1



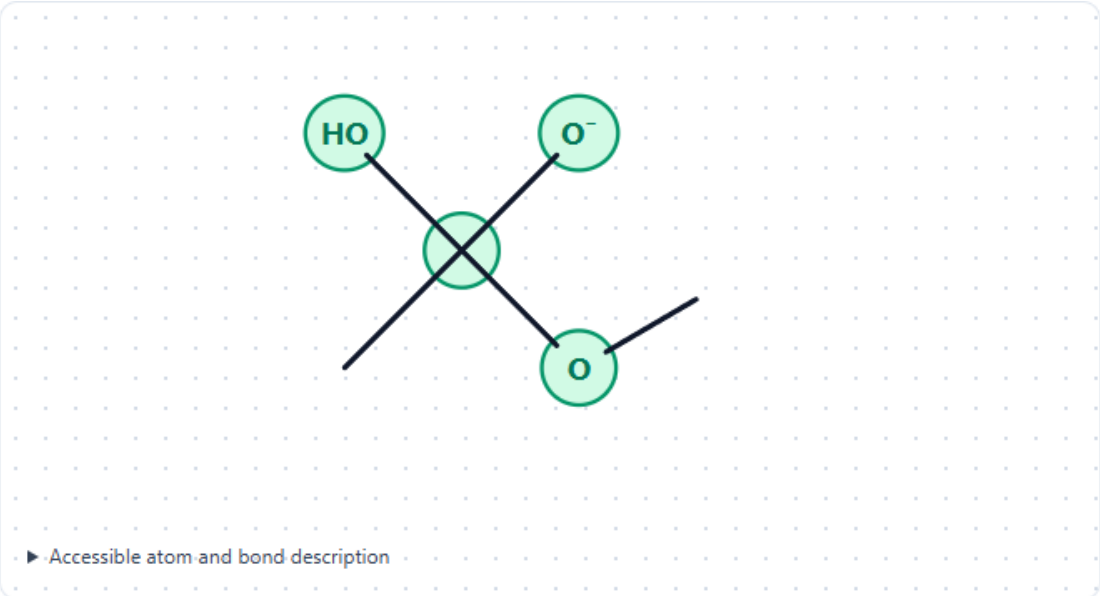
acyl-mechanism:step-2 | 2. Tetrahedral intermediate

The carbon now bears the incoming OH, the original OCH3, and an O− derived from the carbonyl oxygen. It has four single bonds. Four directions are spread in a 2D projection for readability, not a claim of 90-degree physical bond angles.

Displayed molecule ID: ester-intermediate

{"highlights": {"prin": [1, 2, 3, 5]}, "revealName": true}

Screenshot: acyl-mechanism-step-2



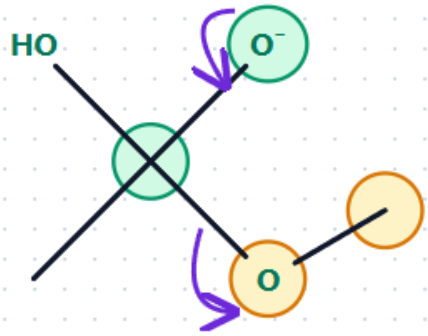
acyl-mechanism:step-3 | 3. Collapse

An oxygen lone pair reforms C=O, and the C–OCH3 bond pair moves to the leaving oxygen. Subsequent proton transfer yields carboxylate plus methanol under basic conditions.

Displayed molecule ID: ester-intermediate

{"highlights": {"prin": [1, 2], "sub": [3, 4], "flow": [{"fromAtom": 2, "toBond": 1}, {"fromBond": 2, "toAtom": 3}]}, "revealName": false}

Screenshot: acyl-mechanism-step-3



▶ Accessible atom and bond description

derivative-reactions | Hydrolysis, alcoholysis, and transesterification

Conditions determine the protonation state

derivative-reactions:rule | Core Rules

Anhydrides react with water to give acids, with alcohols to give ester plus acid, and with amines to give amide plus acid-derived products. Transesterification exchanges one OR group for another under suitable catalysis. Acidic amide hydrolysis gives carboxylic acid and protonated amine; basic hydrolysis gives carboxylate and amine. A condensation joins reactants while losing a small molecule. Fischer esterification is acid + alcohol ⇌ ester + water under acid catalysis; removing water or using excess alcohol favors ester. Water need not be completely absent, and aqueous acid promotes the reverse hydrolysis.

derivative-reactions:cue | Memory Cues

Follow the acyl fragment and the departing group separately. Saponification is basic ester cleavage; a triacylglycerol yields glycerol and fatty-acid salts. Suitable dicarboxylic acids can dehydrate on heating to five- or six-membered cyclic anhydrides when geometry and conditions allow; not every diacid cyclizes.

derivative-reactions:trap | Common Mistakes / MCAT Traps

In basic ester hydrolysis, carboxylate formation helps drive the net reaction. Do not draw a free carboxylic acid as the major product until acid workup. Amide hydrolysis usually needs more forcing conditions than ester hydrolysis. Acetal/ketal protection masks an aldehyde or ketone electronically and can tolerate many bases or hydrides; aqueous acid removes it. It does not make a carbonyl universally inert or merely block approach by steric bulk.

derivative-reactions:app | MCAT Applications

Proteases catalyze peptide hydrolysis under mild biological conditions by activating nucleophiles and stabilizing intermediates; the uncatalyzed reaction is much slower.

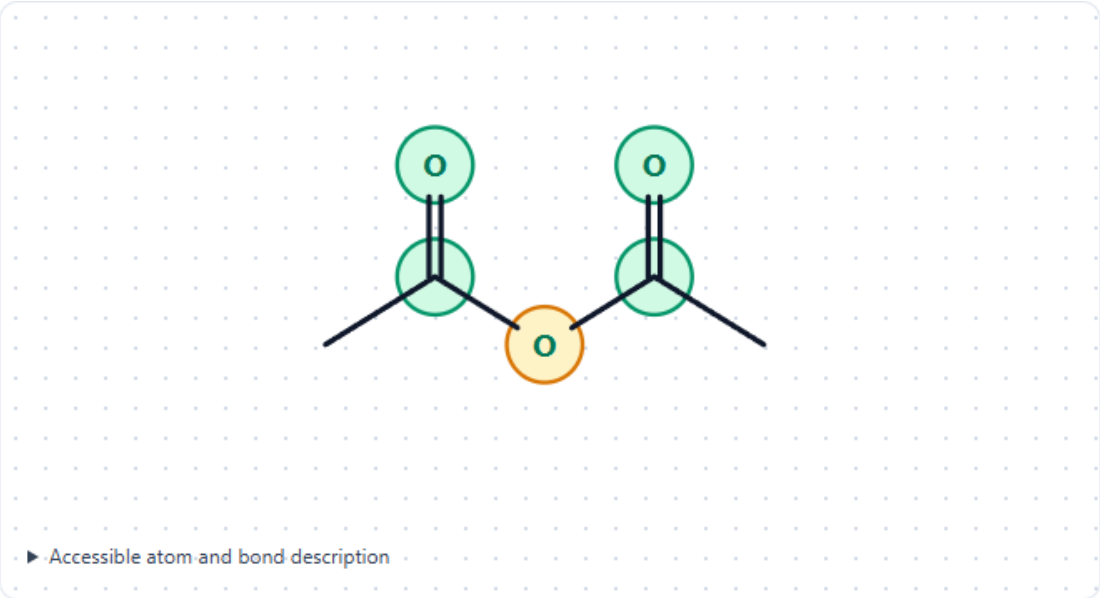
derivative-reactions:step-1 | 1. Anhydride cleavage

Water attack at either equivalent acyl carbon of ethanoic anhydride leads overall to two ethanoic acid molecules.

Displayed molecule ID: anhyd

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Screenshot: derivative-reactions-step-1



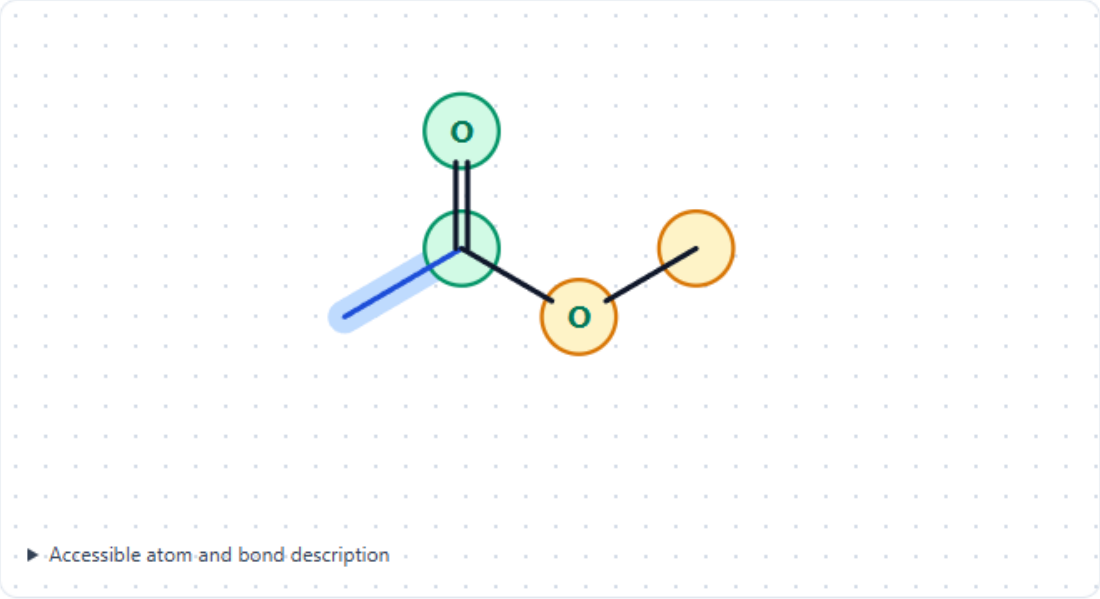
derivative-reactions:step-2 | 2. Exchange an ester group

In transesterification, the O-alkyl group can be replaced by the incoming alcohol-derived group while the acyl carbon skeleton remains.

Displayed molecule ID: methylacetate

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Screenshot: derivative-reactions-step-2



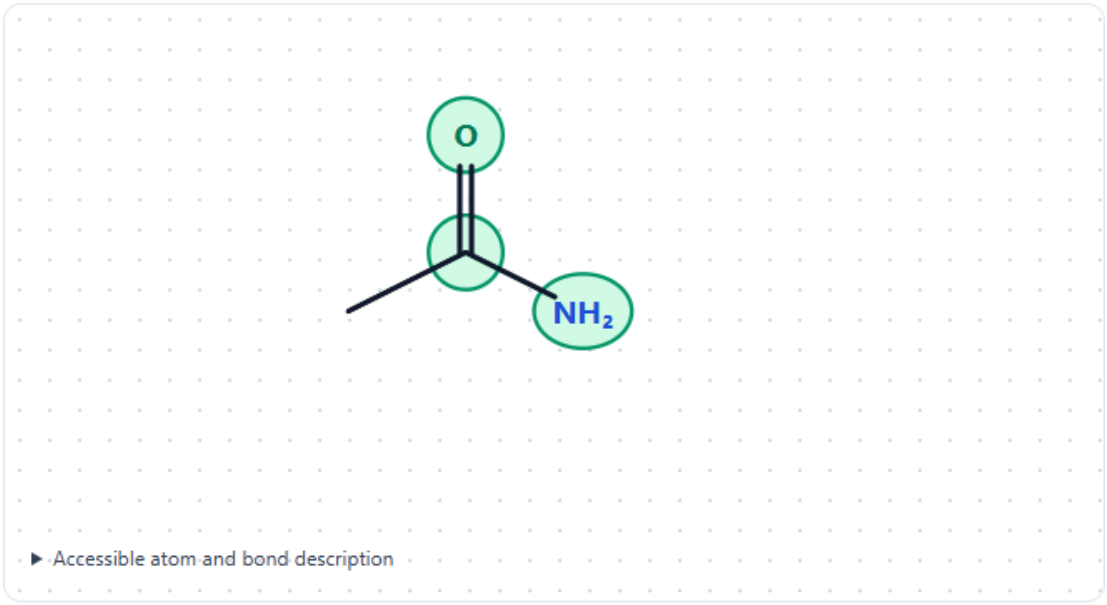
derivative-reactions:step-3 | 3. Hydrolyze an amide

An amide's N donation lowers electrophilicity and its leaving group is poor unless conditions enable departure. Strong acid/base with heating or a catalyst can accomplish hydrolysis.

Displayed molecule ID: acetamide

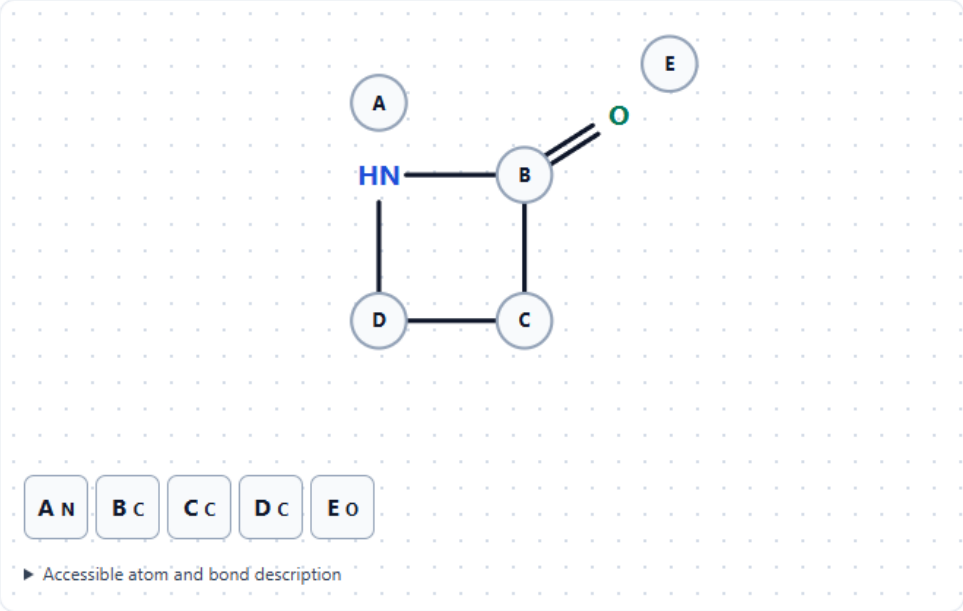
{"highlights": {"prin": [1, 2, 3]}, "revealName": true}

Screenshot: derivative-reactions-step-3



cyclic-derivative | Identify a strained derivative

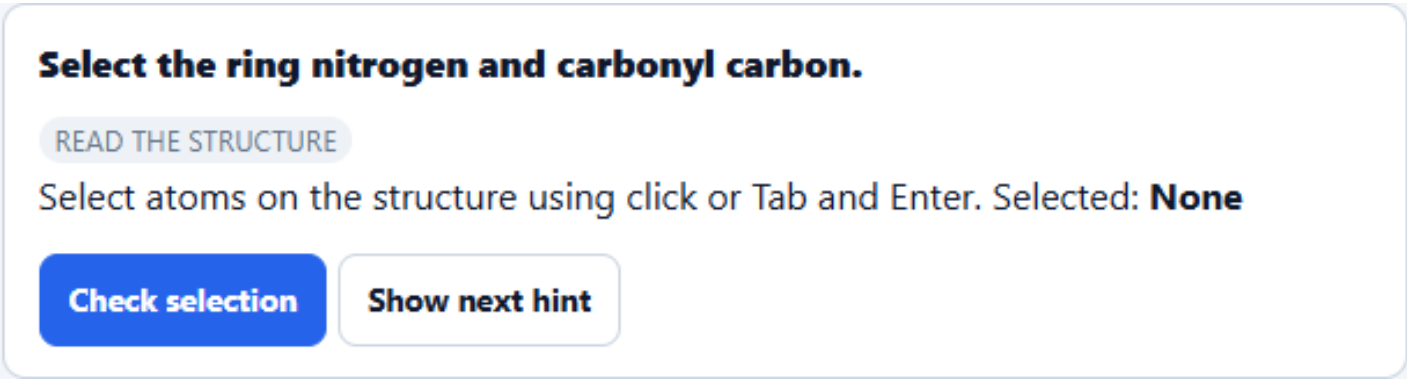
Screenshot: cyclic-derivative-stage-1-molecule



cyclic-derivative:stage-1 | Select the ring nitrogen and carbonyl carbon.

Answer type: atoms

Screenshot: cyclic-derivative-stage-1-question



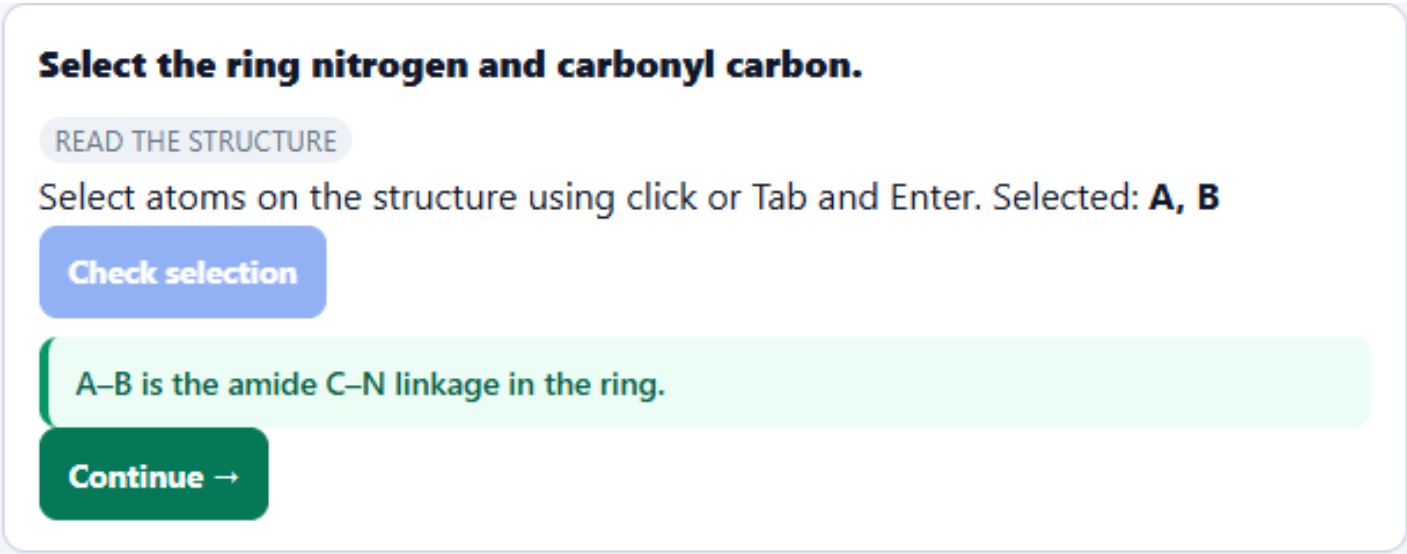
{"authorAnswer": [0, 1], "tolerance": null, "highlightsAfterCorrectAnswer": {"prin": [0, 1]}}

Hint 1: Find N bonded directly to C=O carbon.

Hint 2: Select both endpoints of that bond.

Explanation: A–B is the amide C–N linkage in the ring.

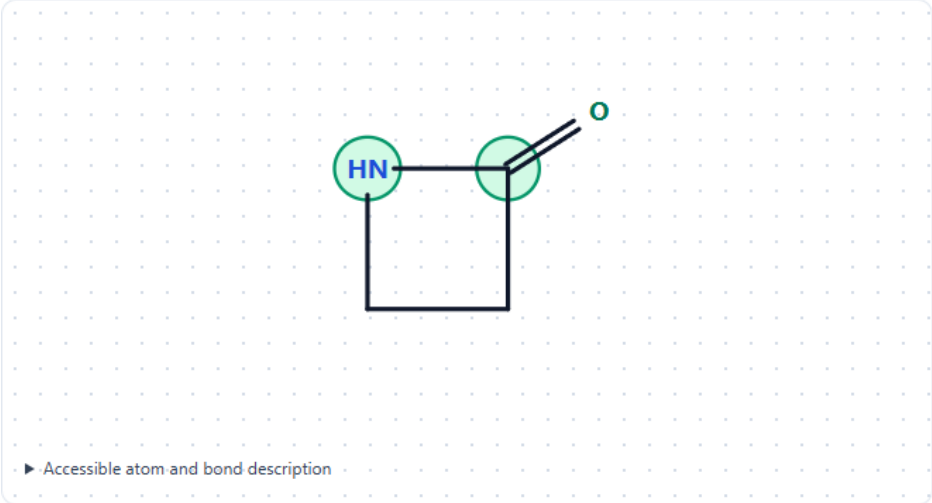
Screenshot: cyclic-derivative-stage-1-feedback



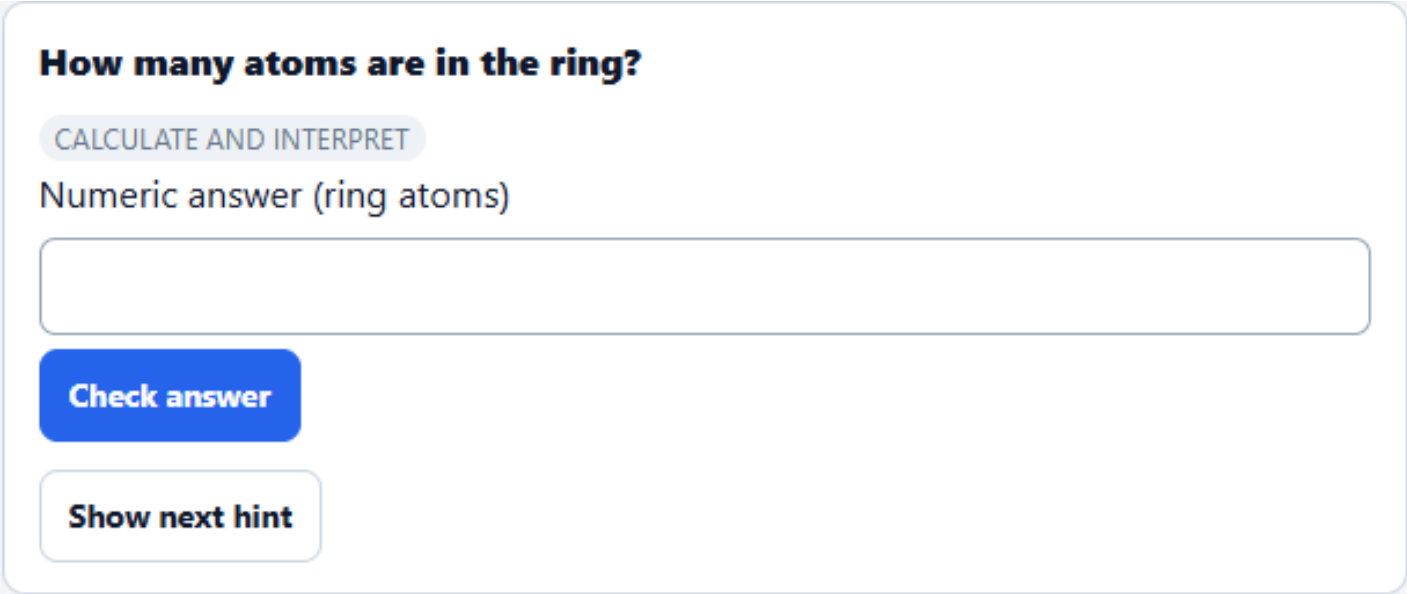
cyclic-derivative:stage-2 | How many atoms are in the ring?

Answer type: number | unit: ring atoms

Screenshot: cyclic-derivative-stage-2-molecule



Screenshot: cyclic-derivative-stage-2-question



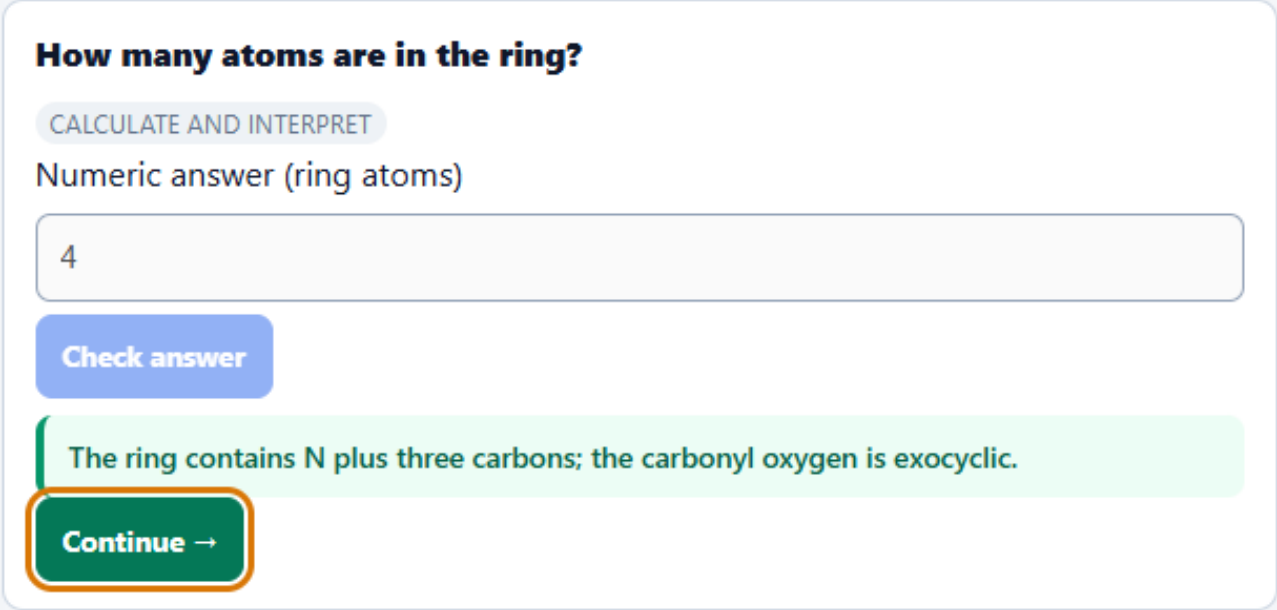
{"authorAnswer": 4, "tolerance": 0, "highlightsAfterCorrectAnswer": {}}

Hint 1: Follow the closed path.

Hint 2: Do not count the oxygen outside the ring.

Explanation: The ring contains N plus three carbons; the carbonyl oxygen is exocyclic.

Screenshot: cyclic-derivative-stage-2-feedback



cyclic-derivative:stage-3 | The functional-group family is...

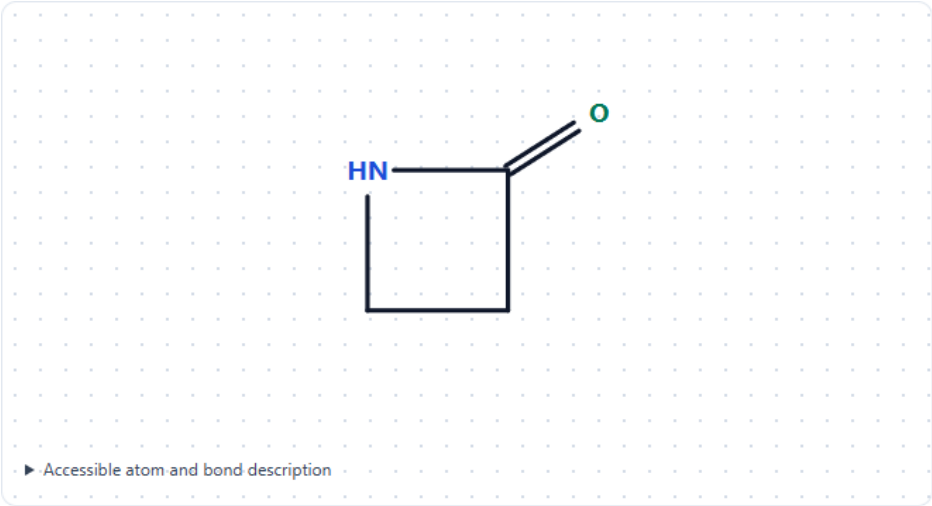
Answer type: choice

Option 0 (A): Lactam

Option 1 (B): Lactone

Option 2 (C): Ether

Screenshot: cyclic-derivative-stage-3-molecule



Screenshot: cyclic-derivative-stage-3-question



{"authorAnswer": 0, "tolerance": null, "highlightsAfterCorrectAnswer": {}}

Hint 1: Use the heteroatom attached to the acyl carbon.

Hint 2: N indicates an amide.

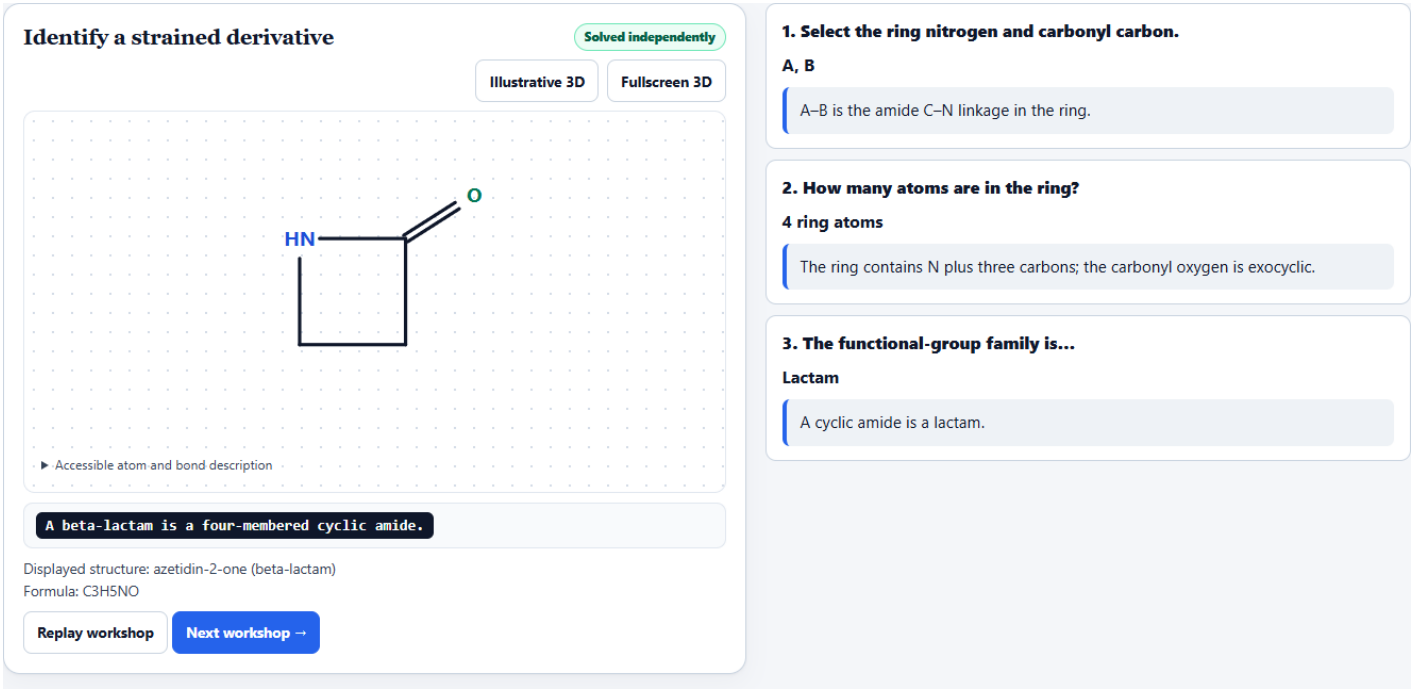
Explanation: A cyclic amide is a lactam.

Screenshot: cyclic-derivative-stage-3-feedback



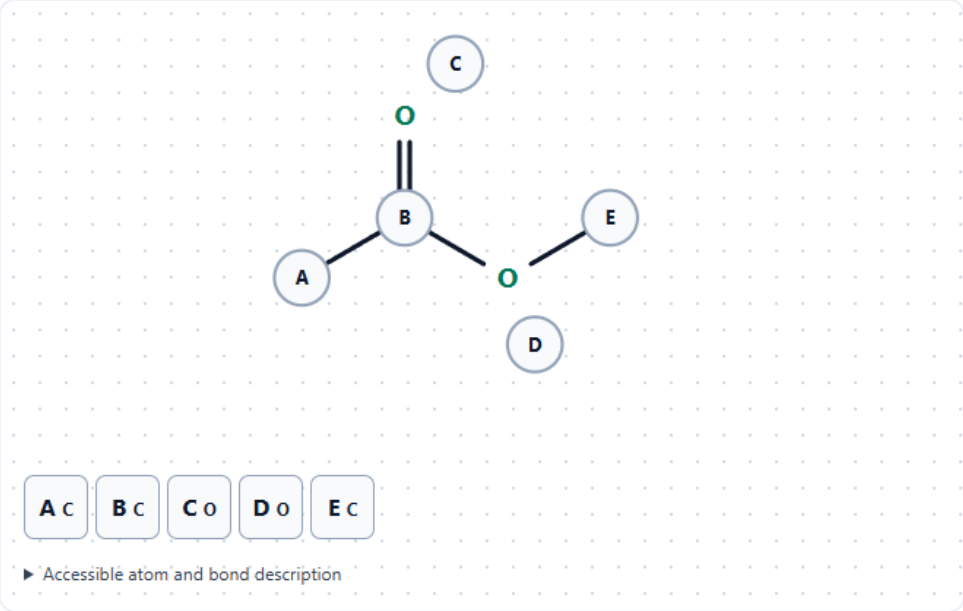
Completion summary: A beta-lactam is a four-membered cyclic amide.

Screenshot: cyclic-derivative-complete



acyl-order | Follow addition–elimination

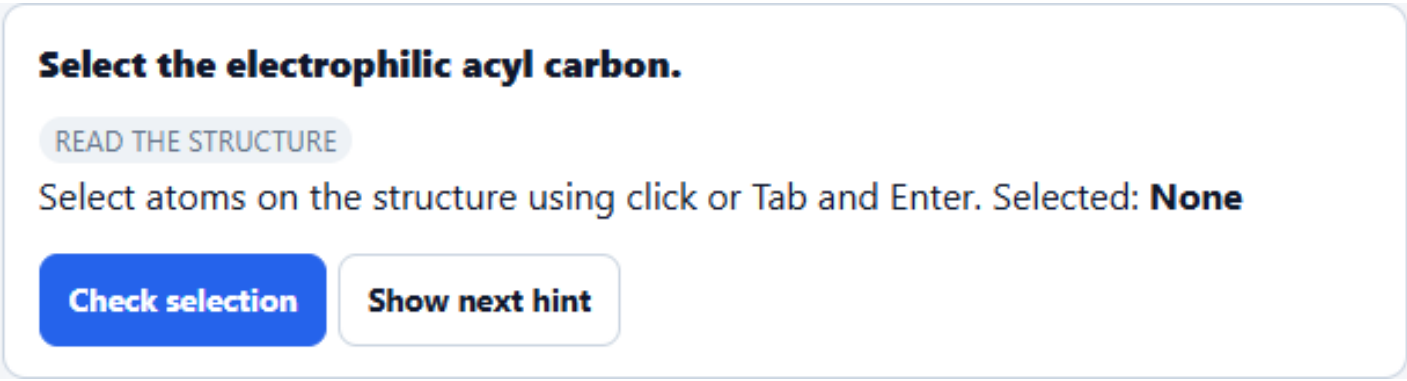
Screenshot: acyl-order-stage-1-molecule



acyl-order:stage-1 | Select the electrophilic acyl carbon.

Answer type: atoms

Screenshot: acyl-order-stage-1-question



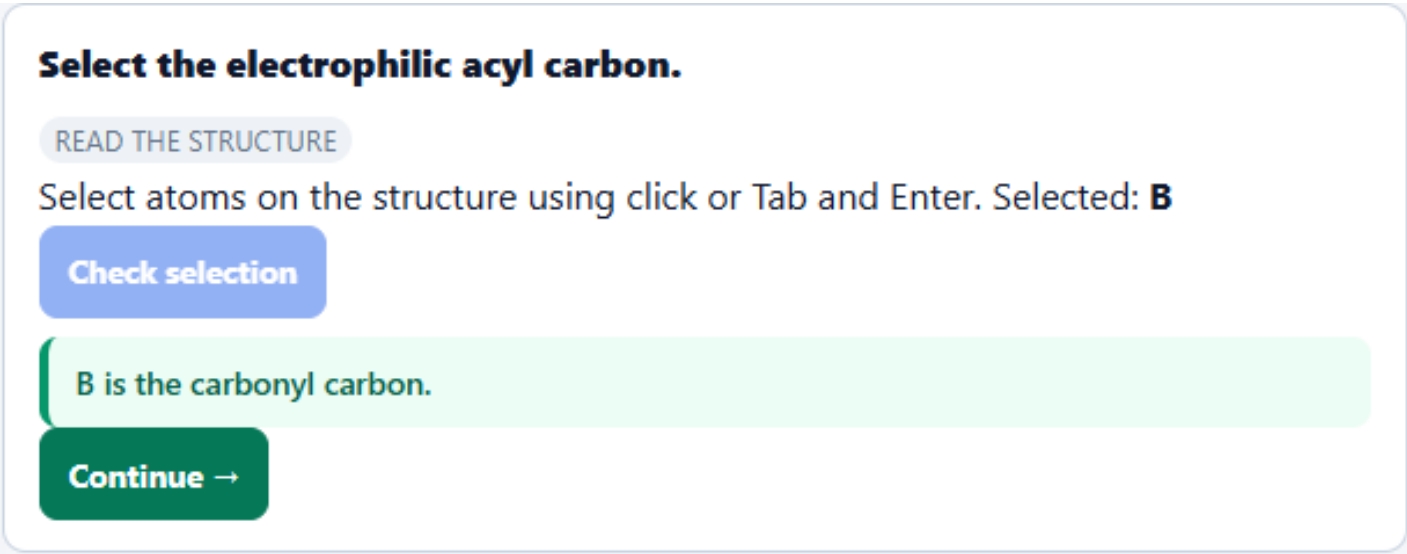
{"authorAnswer": [1], "tolerance": null, "highlightsAfterCorrectAnswer": {"prin": [1, 2]}}

Hint 1: Find C=O.

Hint 2: Choose carbon, not oxygen.

Explanation: B is the carbonyl carbon.

Screenshot: acyl-order-stage-1-feedback



acyl-order:stage-2 | After addition, what restores the carbonyl?

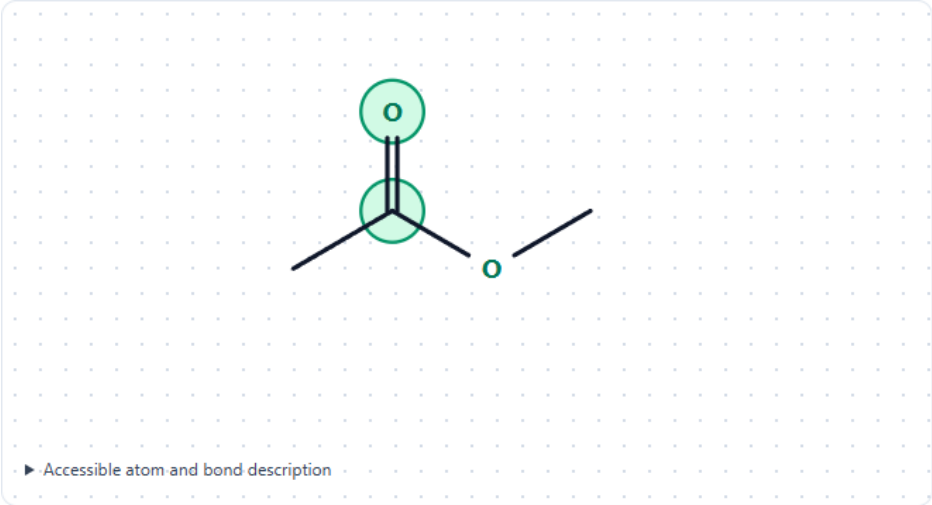
Answer type: choice

Option 0 (A): Collapse of a tetrahedral intermediate with leaving-group departure

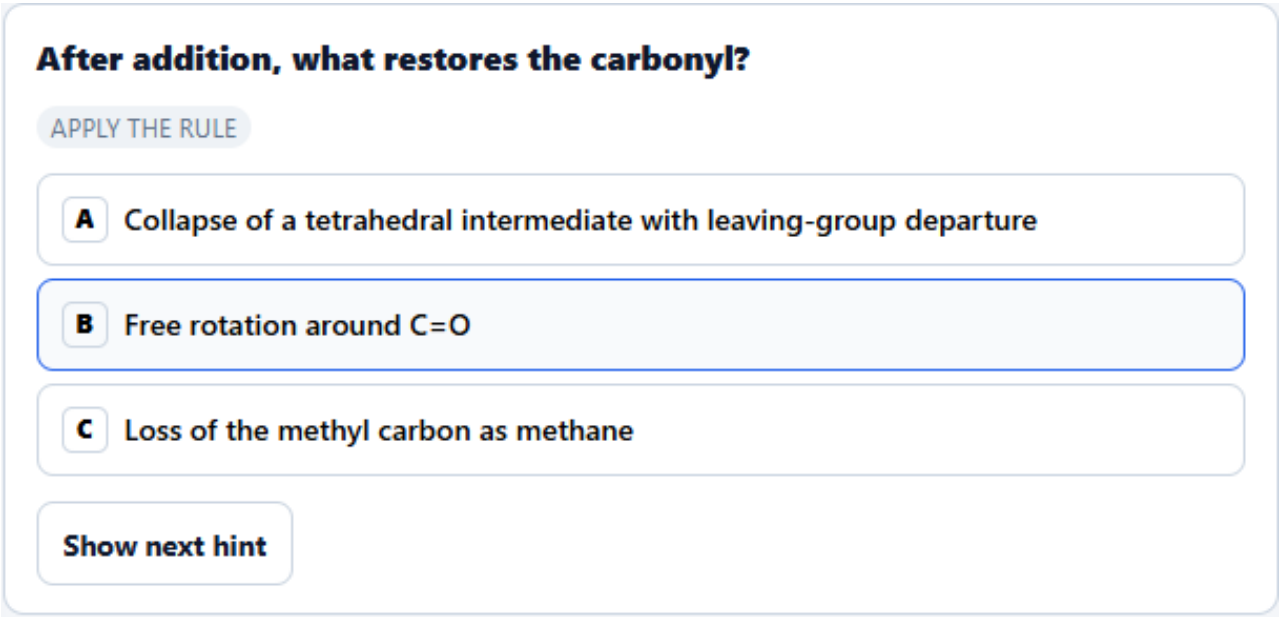
Option 1 (B): Free rotation around C=O

Option 2 (C): Loss of the methyl carbon as methane

Screenshot: acyl-order-stage-2-molecule



Screenshot: acyl-order-stage-2-question



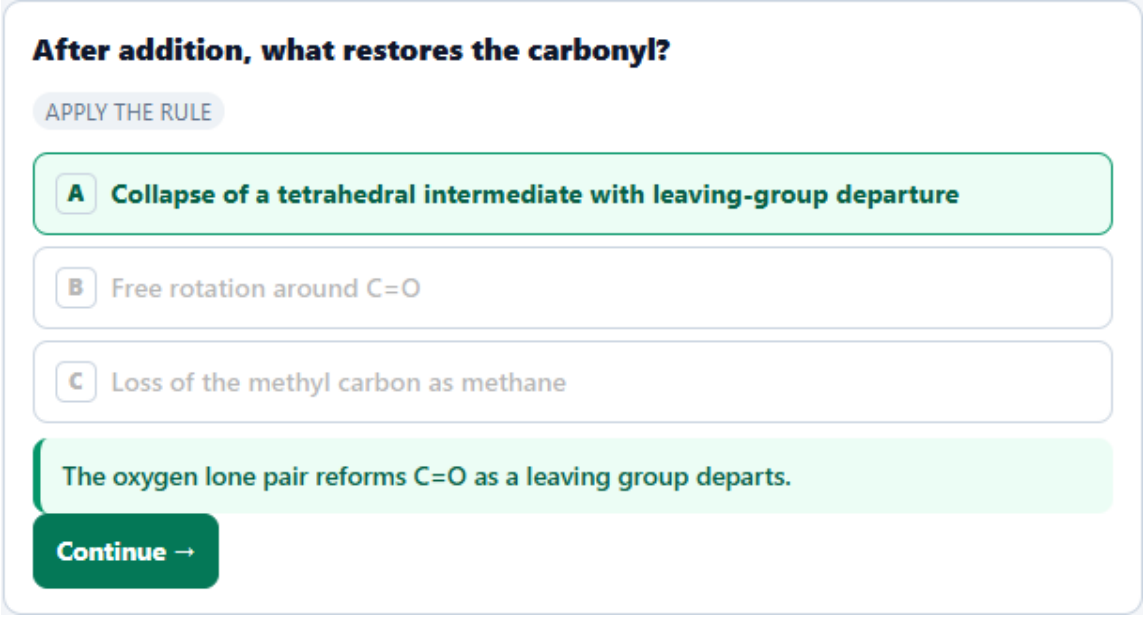
{"authorAnswer": 0, "tolerance": null, "highlightsAfterCorrectAnswer": {}}

Hint 1: The acyl carbon must return to three sigma directions.

Hint 2: Reform the pi bond and expel the heteroatom leaving group.

Explanation: The oxygen lone pair reforms C=O as a leaving group departs.

Screenshot: acyl-order-stage-2-feedback



acyl-order:stage-3 | Among comparable ordinary anhydrides, esters, and amides, which is least reactive toward acyl substitution?

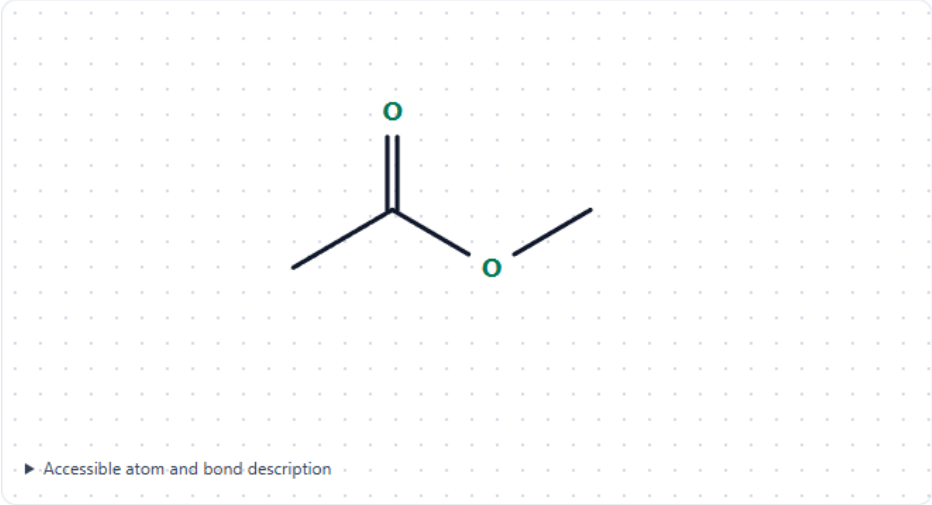
Answer type: choice

Option 0 (A): Anhydride

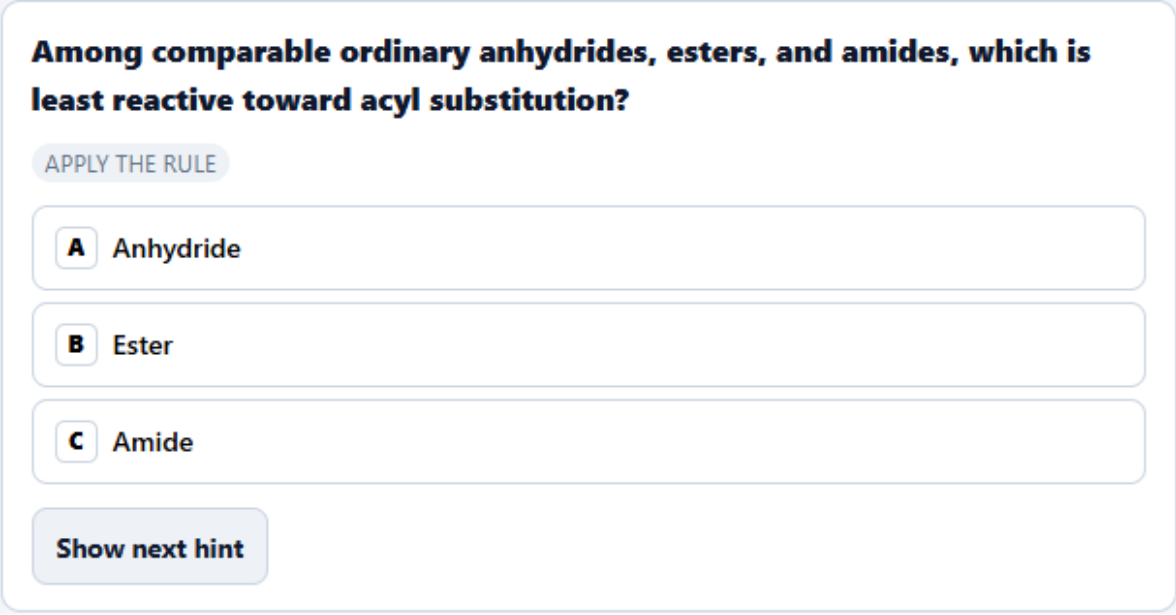
Option 1 (B): Ester

Option 2 (C): Amide

Screenshot: acyl-order-stage-3-molecule



Screenshot: acyl-order-stage-3-question



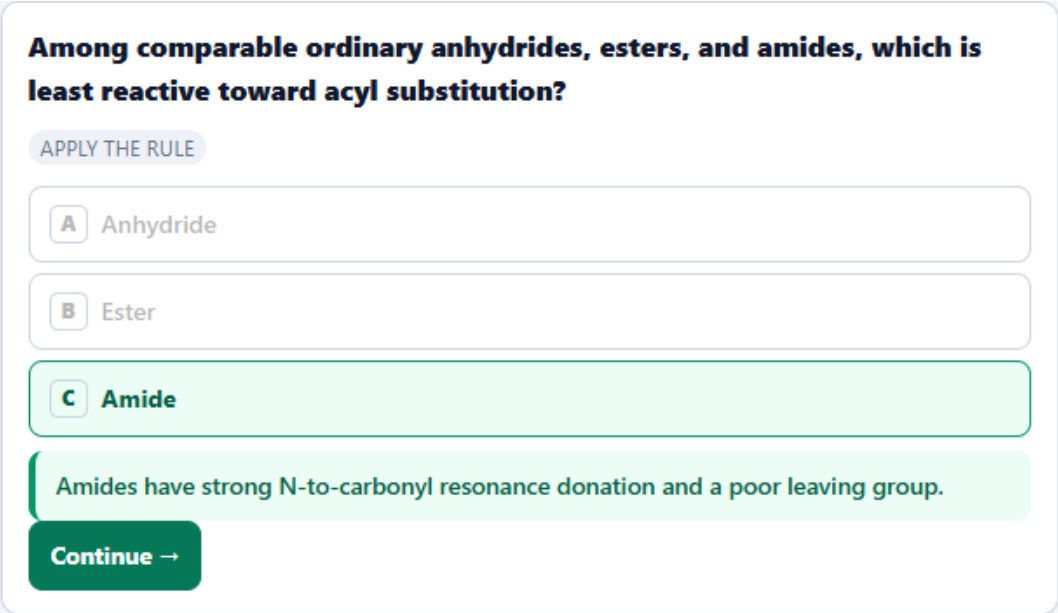
{"authorAnswer": 2, "tolerance": null, "highlightsAfterCorrectAnswer": {}}

Hint 1: Consider lone-pair donation and leaving-group basicity.

Hint 2: Nitrogen strongly stabilizes the amide system.

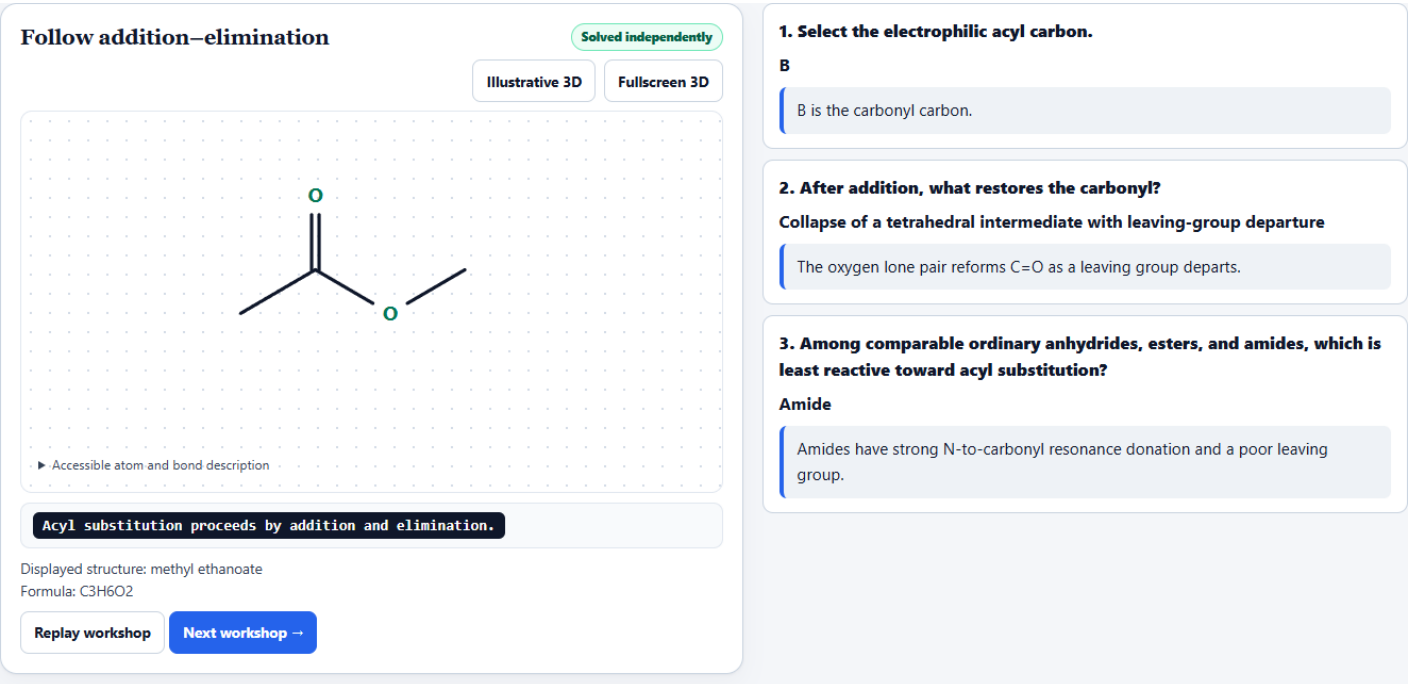
Explanation: Amides have strong N-to-carbonyl resonance donation and a poor leaving group.

Screenshot: acyl-order-stage-3-feedback



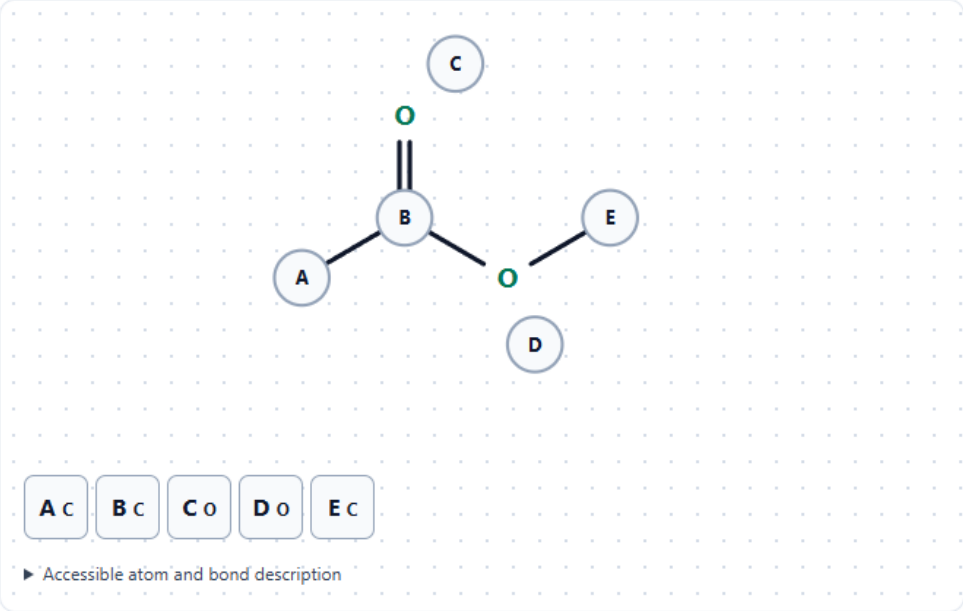
Completion summary: Acyl substitution proceeds by addition and elimination.

Screenshot: acyl-order-complete



hydrolysis-products | Track hydrolysis and workup

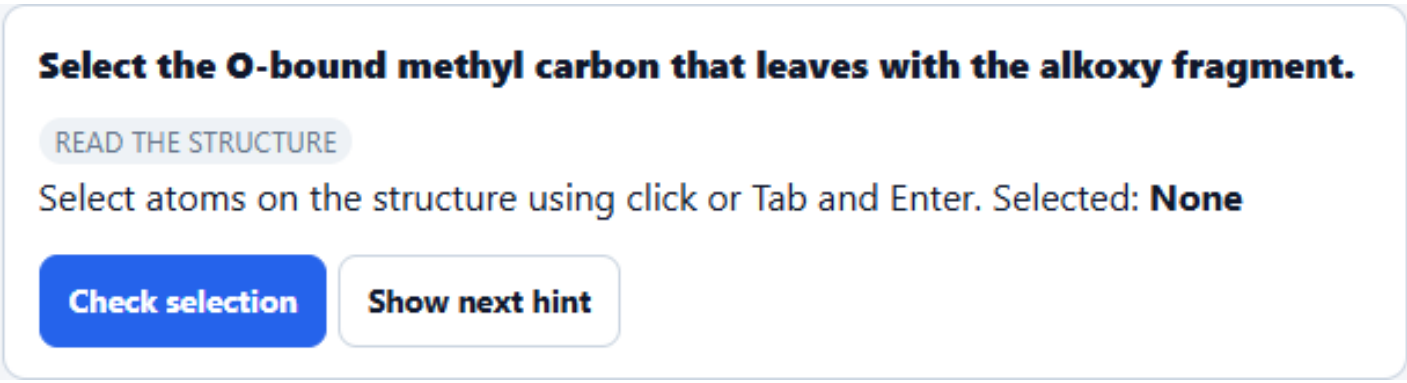
Screenshot: hydrolysis-products-stage-1-molecule



hydrolysis-products:stage-1 | Select the O-bound methyl carbon that leaves with the alkoxy fragment.

Answer type: atoms

Screenshot: hydrolysis-products-stage-1-question



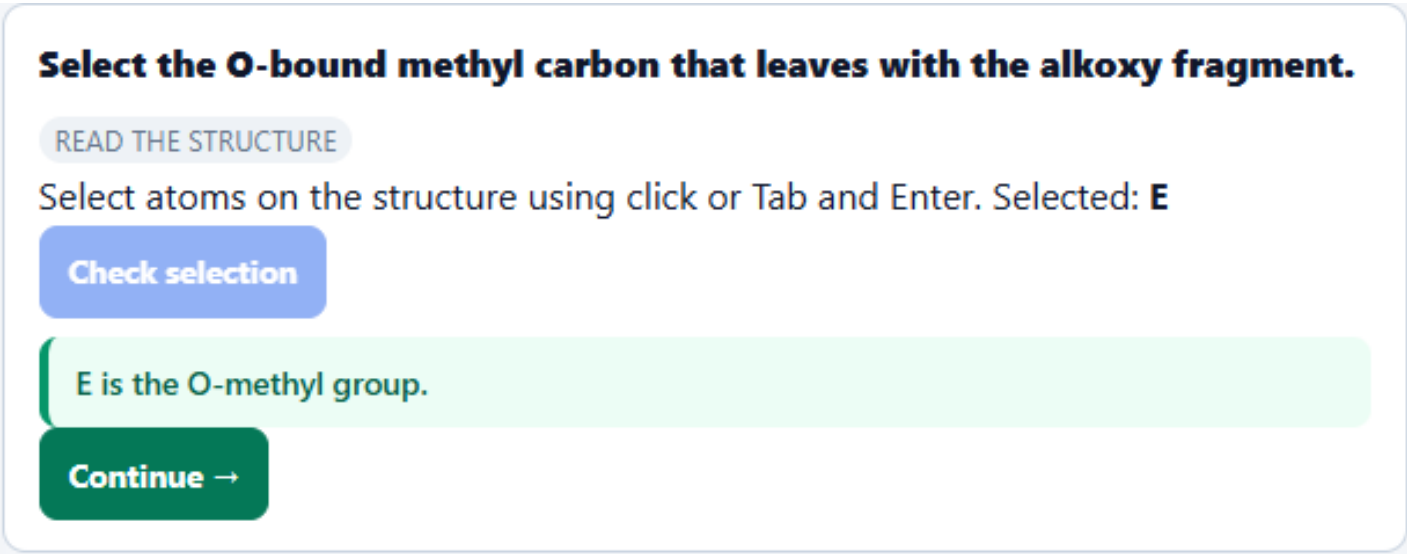
{"authorAnswer": [4], "tolerance": null, "highlightsAfterCorrectAnswer": {"sub": [3, 4]}}

Hint 1: Distinguish the acyl methyl from the O-methyl.

Hint 2: Follow the single-bonded oxygen away from the carbonyl.

Explanation: E is the O-methyl group.

Screenshot: hydrolysis-products-stage-1-feedback



hydrolysis-products:stage-2 | Under excess aqueous NaOH, the major acid-derived product is…

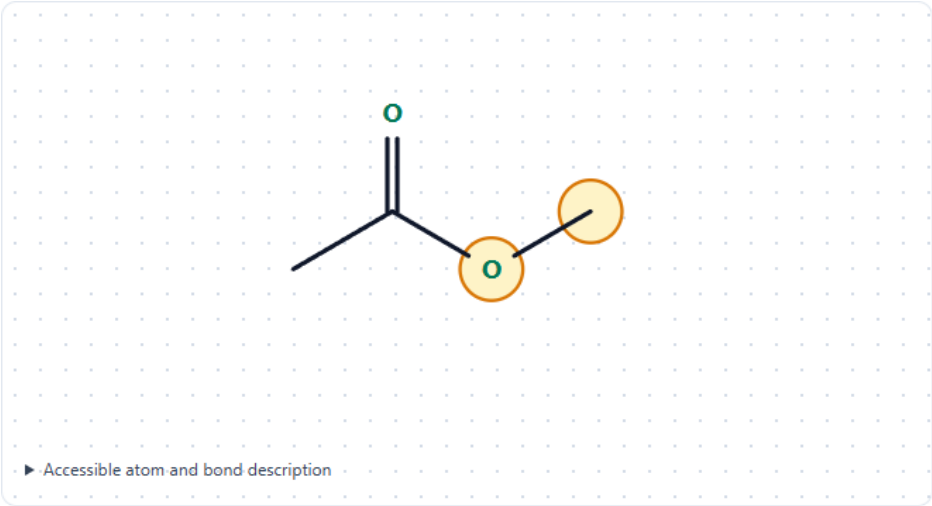
Answer type: choice

Option 0 (A): Acetate (ethanoate)

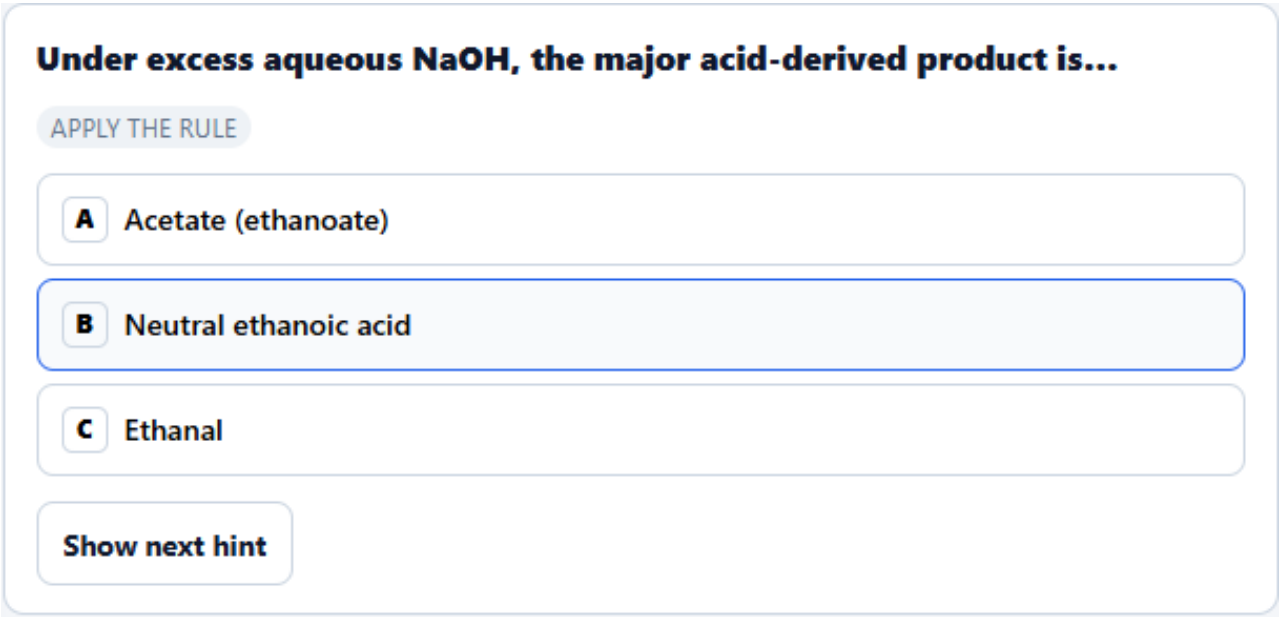
Option 1 (B): Neutral ethanoic acid

Option 2 (C): Ethanal

Screenshot: hydrolysis-products-stage-2-molecule



Screenshot: hydrolysis-products-stage-2-question



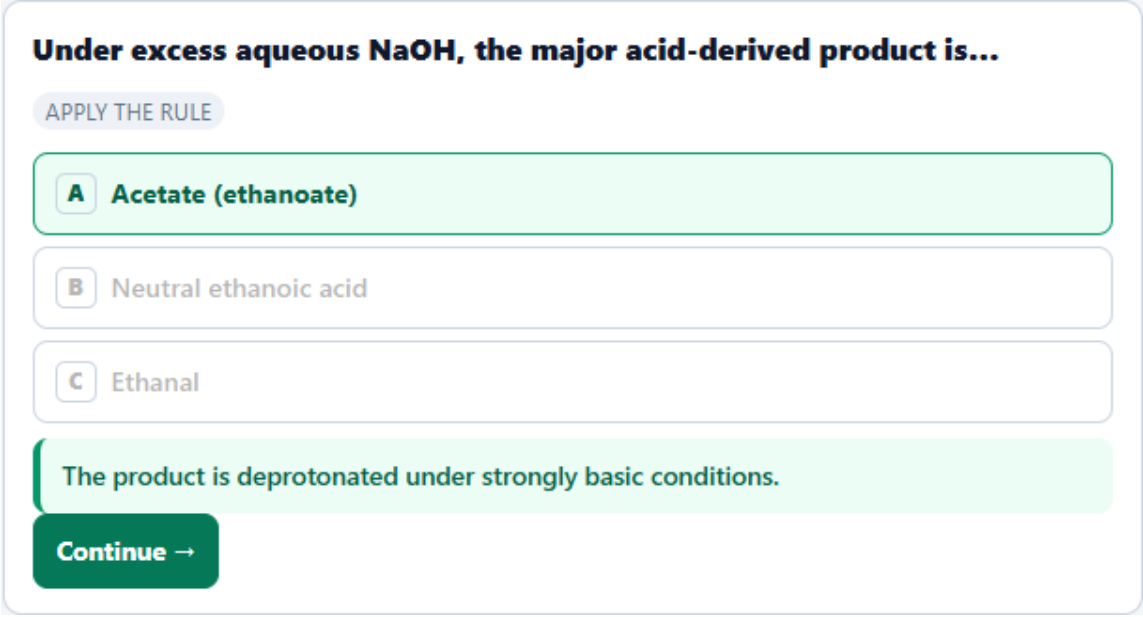
{"authorAnswer": 0, "tolerance": null, "highlightsAfterCorrectAnswer": {}}

Hint 1: Account for acid–base chemistry after substitution.

Hint 2: Carboxylic acids have much lower pKa than water.

Explanation: The product is deprotonated under strongly basic conditions.

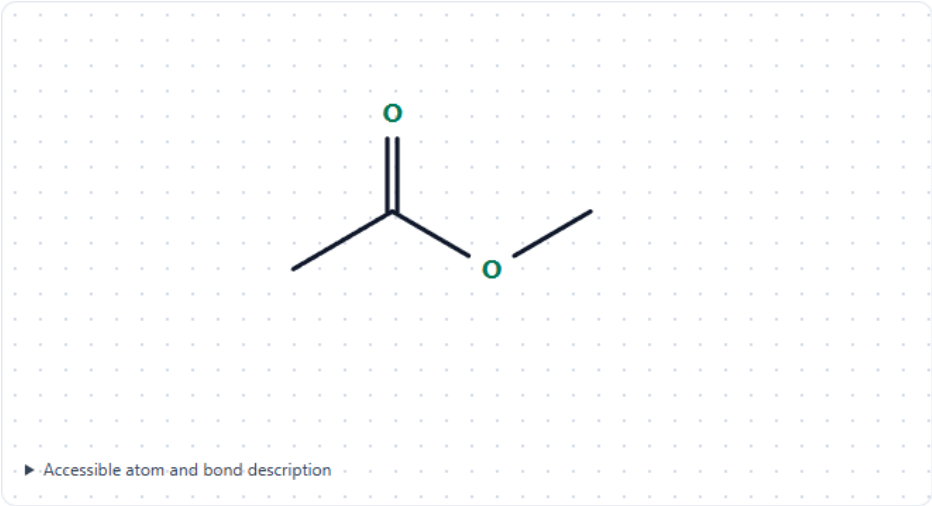
Screenshot: hydrolysis-products-stage-2-feedback



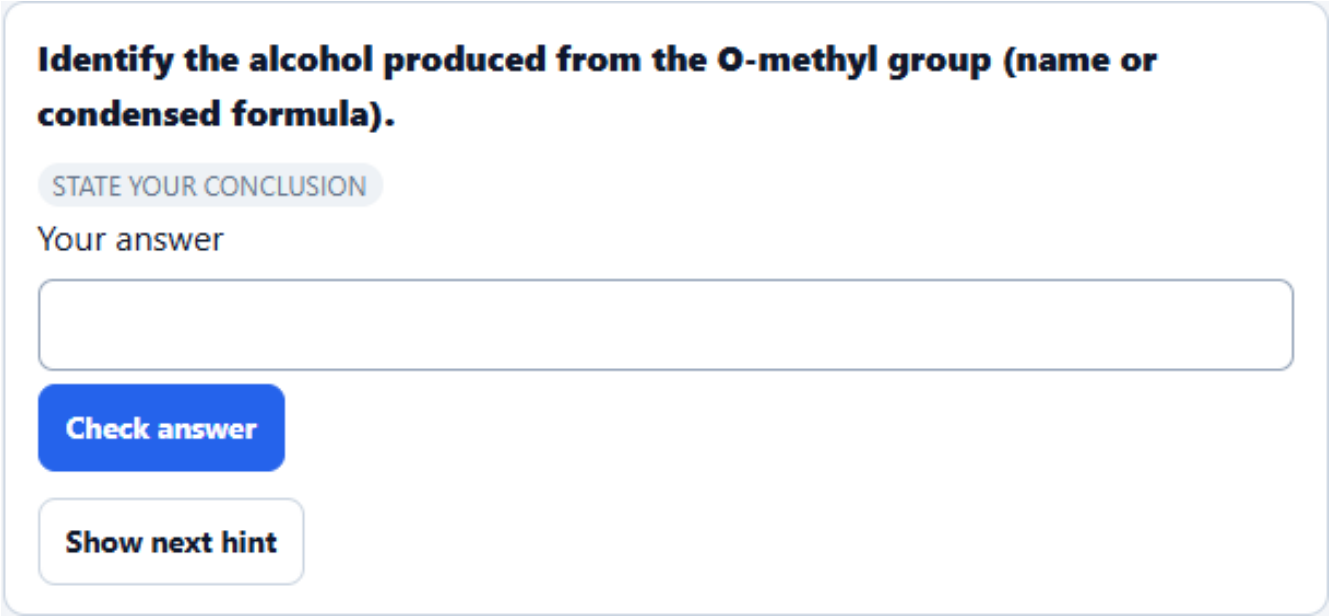
hydrolysis-products:stage-3 | Identify the alcohol produced from the O-methyl group (name or condensed formula).

Answer type: text

Screenshot: hydrolysis-products-stage-3-molecule



Screenshot: hydrolysis-products-stage-3-question



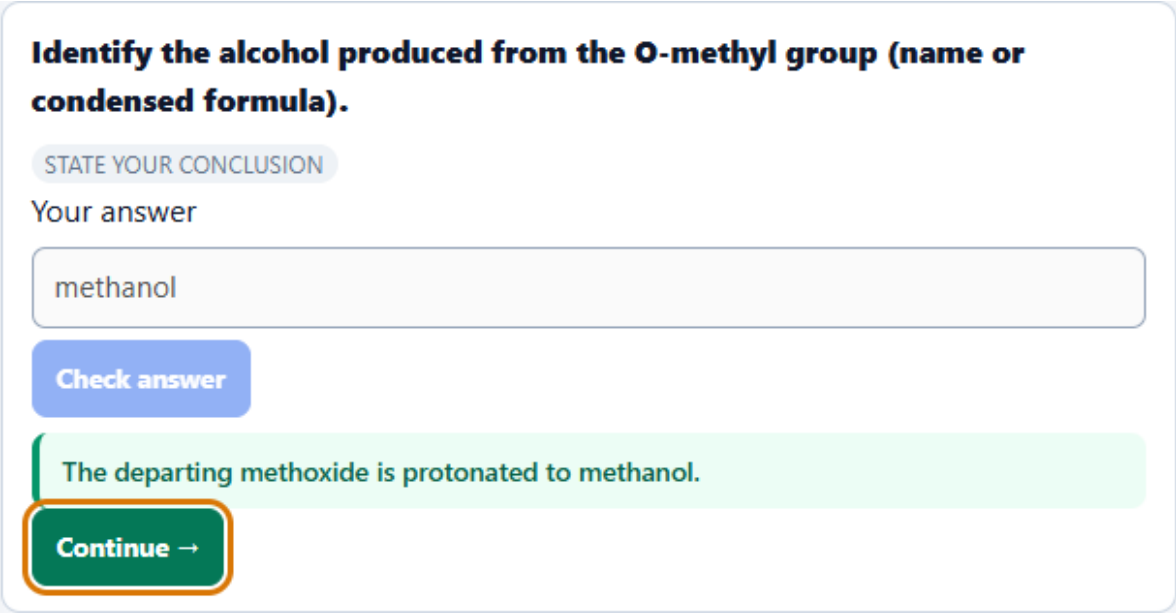
{"authorAnswer": ["methanol", "methyl alcohol", "CH3OH", "CH3-OH", "H3COH", "H3C-OH", "HOCH3", "HO-CH3", "MeOH", "Me-OH", "carbinol"], "tolerance": null, "highlightsAfterCorrectAnswer": {}}

Hint 1: Keep the one-carbon alkoxy fragment.

Hint 2: Replace its bond to acyl carbon with H on oxygen.

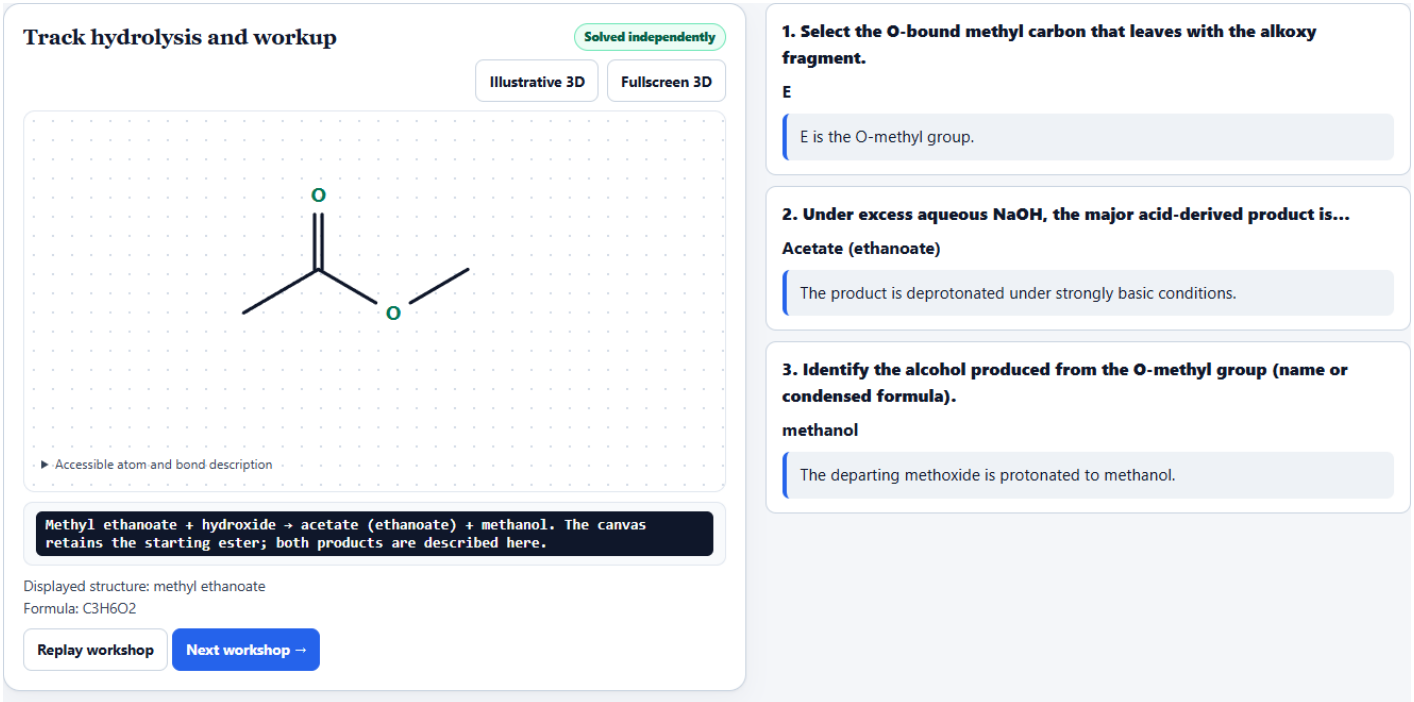
Explanation: The departing methoxide is protonated to methanol.

Screenshot: hydrolysis-products-stage-3-feedback



Completion summary: Methyl ethanoate + hydroxide → acetate (ethanoate) + methanol. The canvas retains the starting ester; both products are described here.

Screenshot: hydrolysis-products-complete



Molecular repository: author claims to verify

ester

Author claims: ethyl propanoate | formula C5H10O2 | canonical composition C5H10O2 | net charge +0

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=48.00, y=115.00; charge=0; explicit H override=none

1 (B): C; x=102.00, y=85.00; charge=0; explicit H override=none

2 (C): C; x=156.00, y=115.00; charge=0; explicit H override=none

3 (D): O; x=156.00, y=52.00; charge=0; explicit H override=none

4 (E): O; x=208.00, y=142.00; charge=0; explicit H override=none

5 (F): C; x=258.00, y=112.00; charge=0; explicit H override=none

6 (G): C; x=308.00, y=142.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 1], [2, 3, 2], [2, 4, 1], [4, 5, 1], [5, 6, 1]]

betalactam

Author claims: azetidin-2-one (beta-lactam) | formula C3H5NO | canonical composition C3H5NO | net charge +0

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): N; x=100.00, y=60.00; charge=0; explicit H override=none

1 (B): C; x=185.00, y=60.00; charge=0; explicit H override=none

2 (C): C; x=185.00, y=145.00; charge=0; explicit H override=none

3 (D): C; x=100.00, y=145.00; charge=0; explicit H override=none

4 (E): O; x=240.00, y=25.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 1], [2, 3, 1], [3, 0, 1], [1, 4, 2]]

lactone

Author claims: oxolan-2-one (gamma-butyrolactone) | formula C4H6O2 | canonical composition C4H6O2 | net charge +0

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): O; x=135.00, y=35.00; charge=0; explicit H override=none

1 (B): C; x=205.00, y=85.00; charge=0; explicit H override=none

2 (C): C; x=180.00, y=165.00; charge=0; explicit H override=none

3 (D): C; x=90.00, y=165.00; charge=0; explicit H override=none

4 (E): C; x=65.00, y=85.00; charge=0; explicit H override=none

5 (F): O; x=265.00, y=55.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 1], [2, 3, 1], [3, 4, 1], [4, 0, 1], [1, 5, 2]]

methylacetate

Author claims: methyl ethanoate | formula C3H6O2 | canonical composition C3H6O2 | net charge +0

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=55.00, y=120.00; charge=0; explicit H override=none

1 (B): C; x=115.00, y=85.00; charge=0; explicit H override=none

2 (C): O; x=115.00, y=25.00; charge=0; explicit H override=none

3 (D): O; x=175.00, y=120.00; charge=0; explicit H override=none

4 (E): C; x=235.00, y=85.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 2], [1, 3, 1], [3, 4, 1]]

ester-intermediate

Author claims: tetrahedral intermediate of methyl ethanoate hydroxide addition | formula C3H7O3 | canonical composition C3H7O3 | net charge -1

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=55.00, y=145.00; charge=0; explicit H override=none

1 (B): C; x=115.00, y=85.00; charge=0; explicit H override=none

2 (C): O; x=175.00, y=25.00; charge=-1; explicit H override=none

3 (D): O; x=175.00, y=145.00; charge=0; explicit H override=none

4 (E): C; x=235.00, y=110.00; charge=0; explicit H override=none

5 (F): O; x=55.00, y=25.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 1], [1, 3, 1], [3, 4, 1], [1, 5, 1]]

anhyd

Author claims: ethanoic anhydride | formula C4H6O3 | canonical composition C4H6O3 | net charge +0

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=45.00, y=130.00; charge=0; explicit H override=none

1 (B): C; x=100.00, y=96.00; charge=0; explicit H override=none

2 (C): O; x=155.00, y=130.00; charge=0; explicit H override=none

3 (D): C; x=210.00, y=96.00; charge=0; explicit H override=none

4 (E): C; x=265.00, y=130.00; charge=0; explicit H override=none

5 (F): O; x=100.00, y=40.00; charge=0; explicit H override=none

6 (G): O; x=210.00, y=40.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 1], [2, 3, 1], [3, 4, 1], [1, 5, 2], [3, 6, 2]]

acetamide

Author claims: ethanamide | formula C2H5NO | canonical composition C2H5NO | net charge +0

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=55.00, y=115.00; charge=0; explicit H override=none

1 (B): C; x=115.00, y=85.00; charge=0; explicit H override=none

2 (C): O; x=115.00, y=25.00; charge=0; explicit H override=none

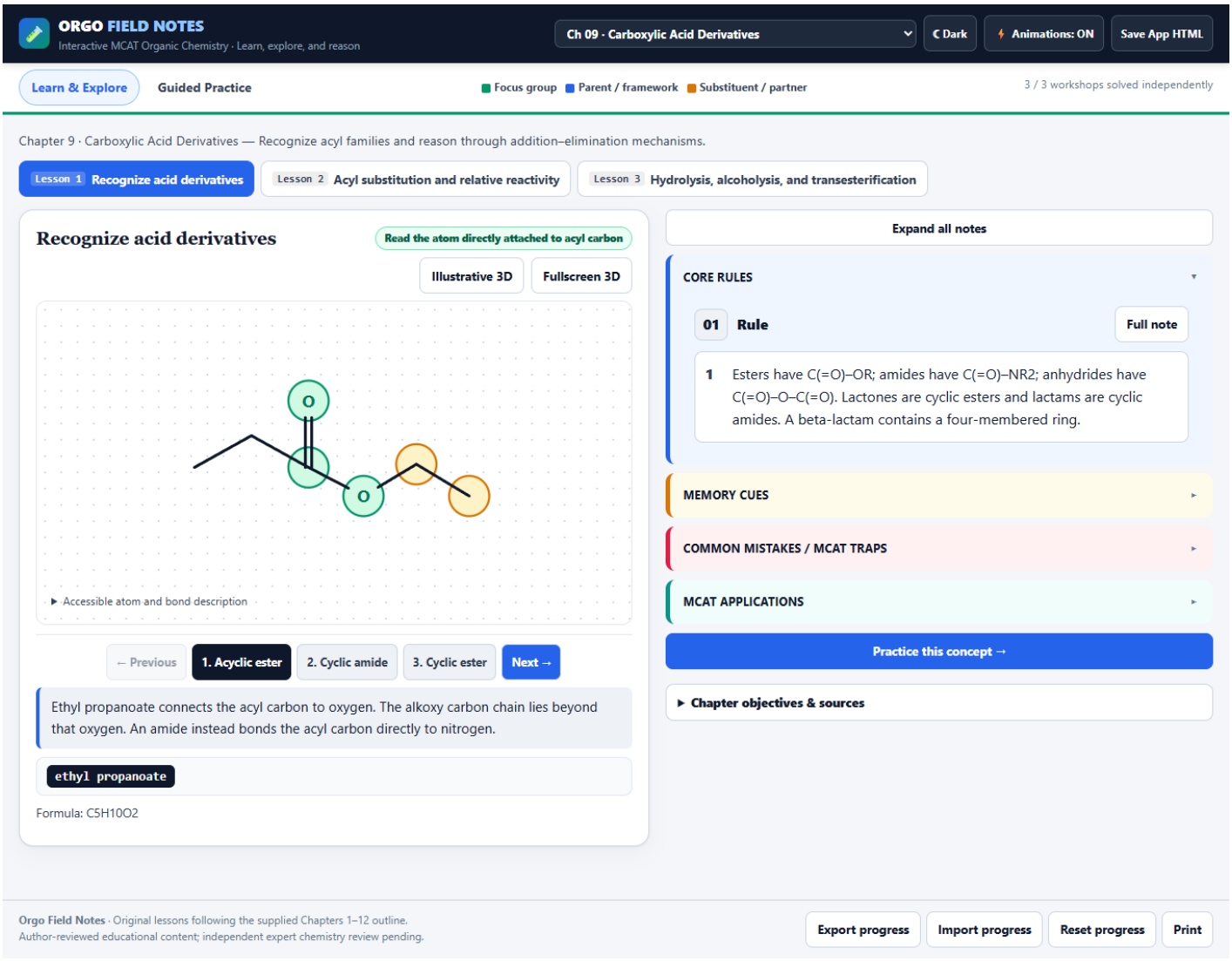
3 (D): N; x=175.00, y=115.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 2], [1, 3, 1]]

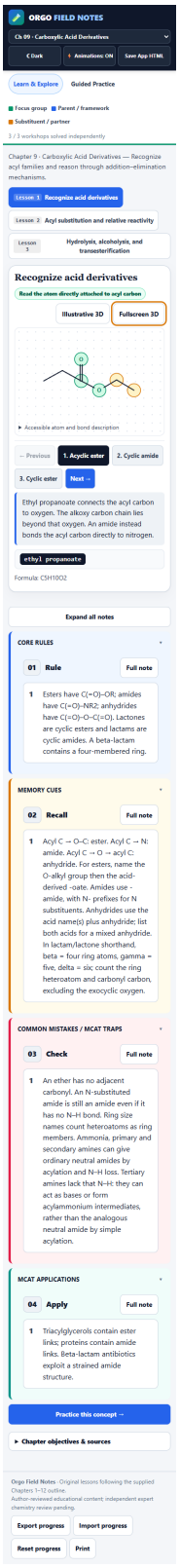
Layout samples

These overview images are scaled for page fit. Detailed structure and stage images appear earlier. Samples cover the first concept on desktop and mobile in both themes, not every responsive state.

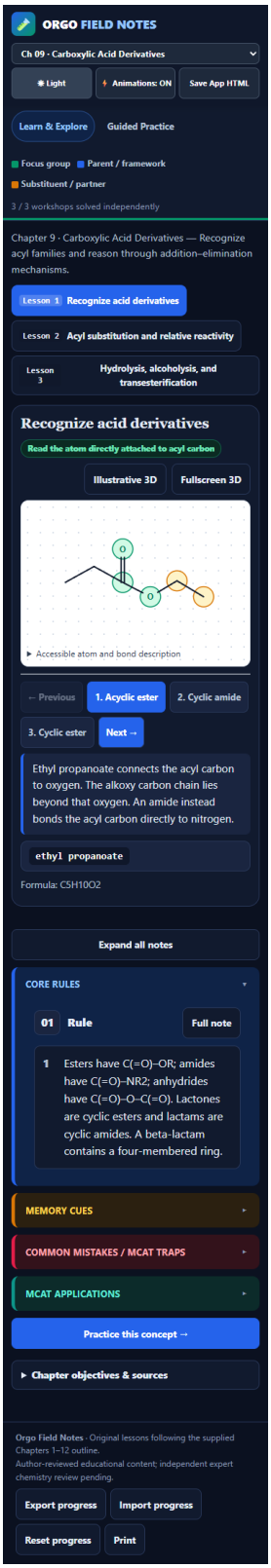
Screenshot: layout-desktop-light



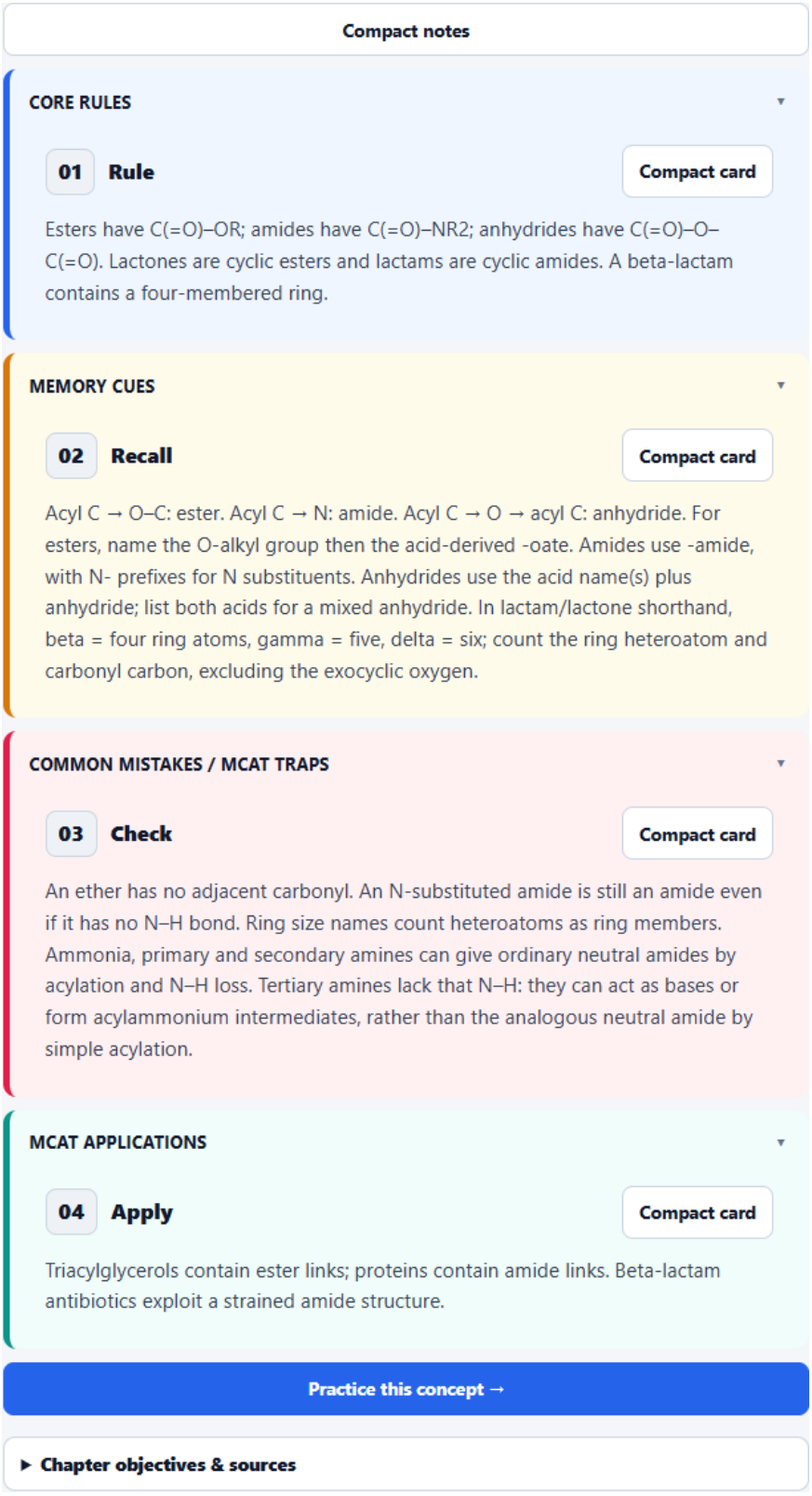
Screenshot: layout-mobile-light



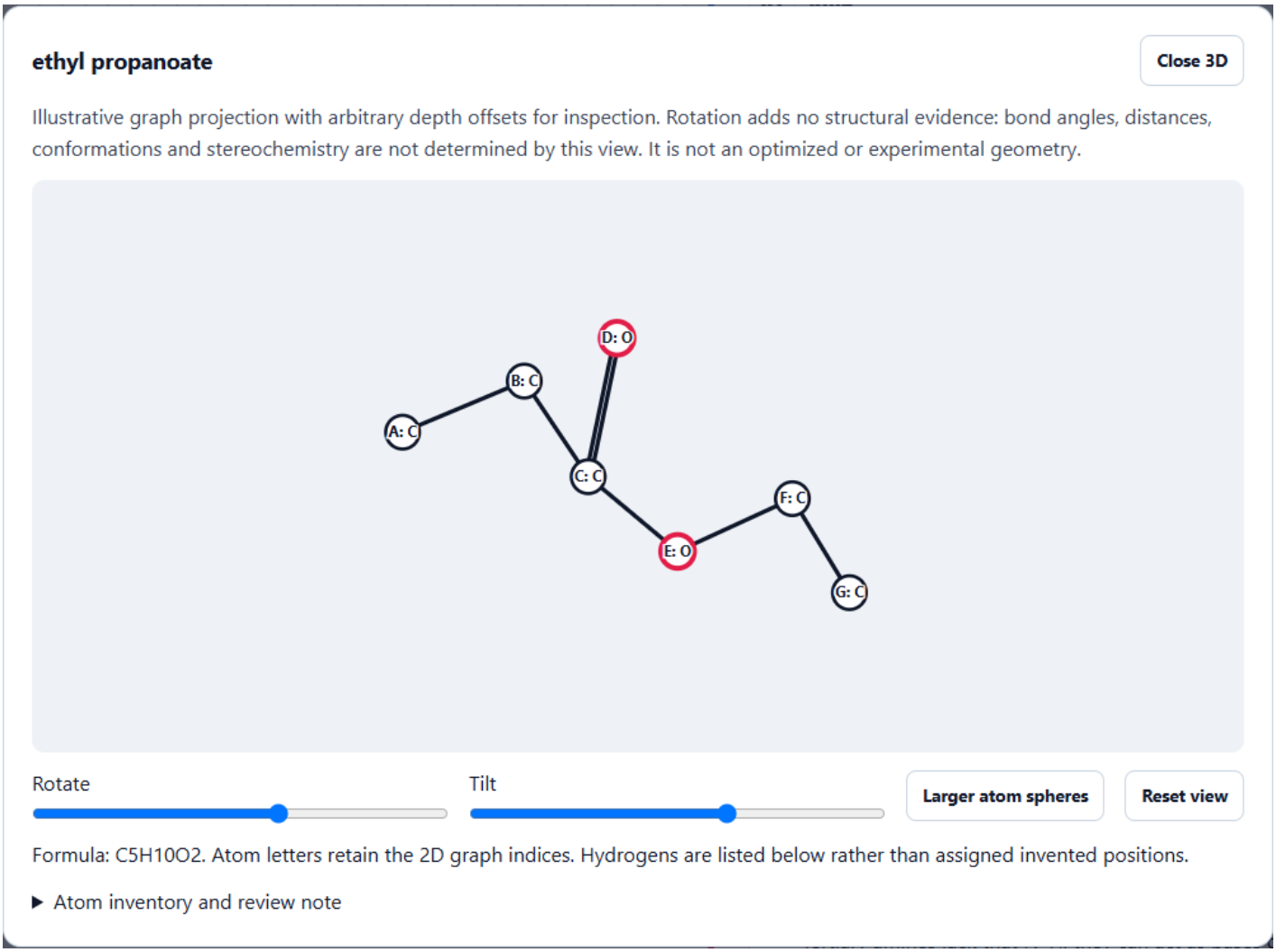
Screenshot: layout-mobile-dark



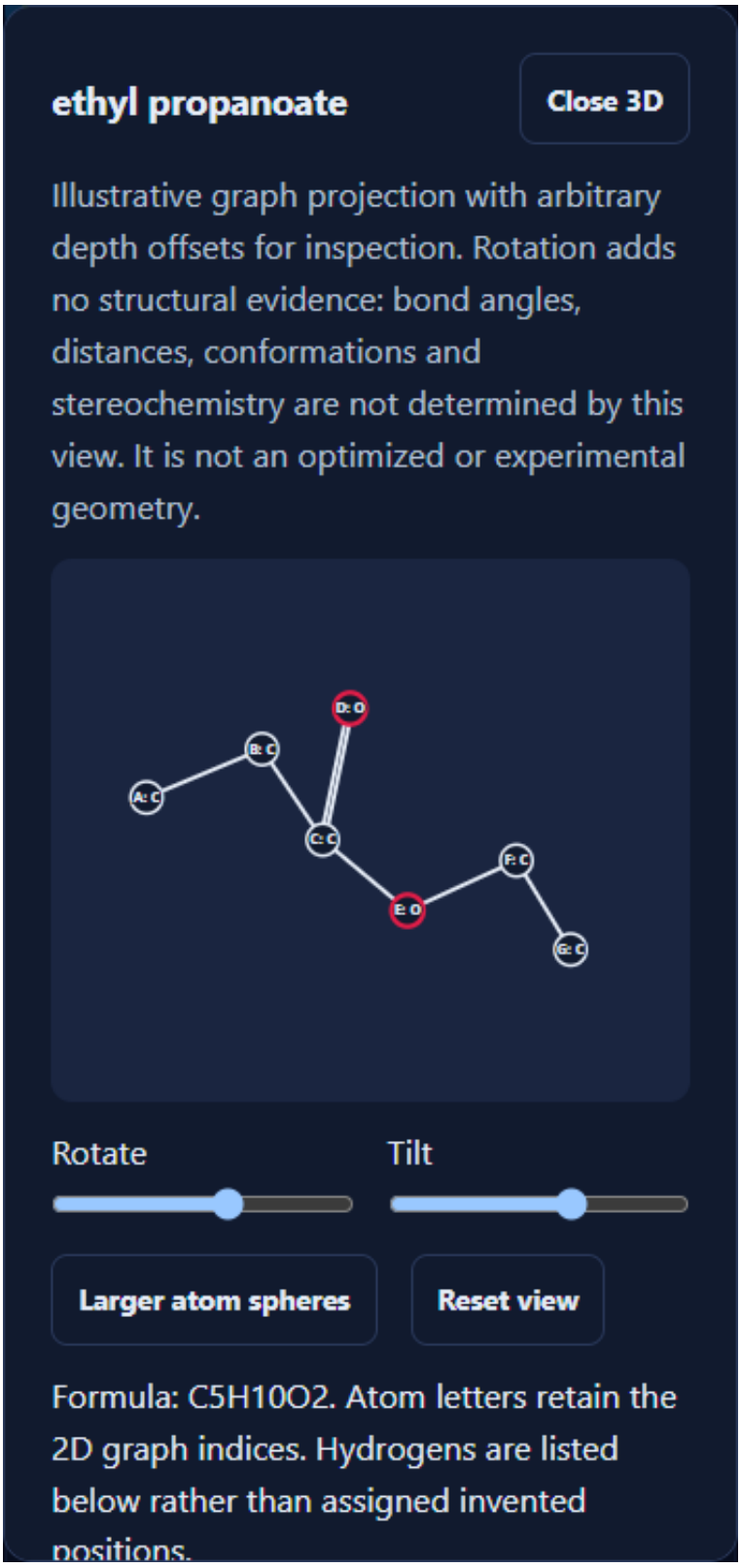
Screenshot: visual-notes-expanded



Screenshot: projection-desktop-light



Screenshot: projection-mobile-dark



Execution evidence

45 screenshots and 9 recorded interactions. Incorrect answers rejected and encoded keys accepted for captured guided stages. This checks implementation against its data; it does not validate chemistry independently. Static screenshots do not verify animation timing, arbitrary clicks, or screen-reader experience.